Generalized Hierarchical Bayesian Segmentation with Irregular Designs, Multi-Sequence Hierarchies, and Grouped/Latent-Group Designs

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ABSTRACT

We develop a generalized hierarchical Bayesian method for multiple-changepoint segmentation of ordered data observed on regular or irregular designs. For a regular exponential-family observation model with a proper segmentwise conjugate prior and finite normalizing constants, cumulative sufficient statistics determine the marginal likelihood and posterior moments of every candidate segment. These segment marginal likelihoods enter dynamic-programming recursions for the marginal likelihood conditional on the number of segments, the posterior distribution of the segment count, marginal changepoint probabilities, and posterior moments of the piecewise-constant signal. A separate max-sum recursion with backtracking returns the joint maximum-a-posteriori partition. Thus, the partition calculation is common across observation families once the segment marginal likelihoods are specified, while the statistical meaning of each segment remains family-specific.

The hierarchical formulation includes design-dependent partition priors for irregular coordinates, common-changepoint models for multiple conditionally independent sequences, known-group models, and a finite latent-group template model. The latent-group algorithm uses a Jensen minorization and exact responsibility-weighted max-sum updates for the stated finite objective; because its component scores are not normalized sampling densities, it is not interpreted as a finite-mixture likelihood. When a nonconjugate segment integral is replaced by a deterministic approximation, a uniform error bound on reachable segment log marginal likelihoods implies explicit bounds on partition marginal likelihoods and posterior odds.

Executed simulation results include boundary-posterior calibration in a repeated Gaussian experiment (ECE approximately 0.010 and Brier score approximately 0.011), examples with Gaussian, Poisson, Binomial, and Beta-valued observations, 96% hard assignment accuracy in a two-group latent-template experiment, and finite-range timings from 0.0455 seconds at $n = 50$ to 1.0585 seconds at $n = 800$ for $k_{\max} = 20$. Four executed case studies illustrate well-log, array-CGH, equity-volatility, and methylation analyses. Because independent reference changepoints are unavailable, these case studies are descriptive rather than accuracy evaluations. Two additional archived computations used an incompatible comparator axis or an invalid predictive family; they are retained as implementation records and excluded from comparative or predictive conclusions.

Keywords Bayesian changepoint analysis, multiple changepoints, product partition models, exponential families, conjugate priors, hierarchical Bayesian models, dynamic programming, irregular designs, multiple sequences, latent groups

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CONTENTS

1 Introduction

Multiple-changepoint analysis represents an ordered sequence by contiguous regimes within which a parameter is constant or otherwise follows a simpler model. The problem appears in signal processing, genomics, econometrics, environmental monitoring, and machine learning. Given observations $y_{1:n}$, possibly recorded at irregular coordinates $x_1 < \cdots < x_n$, the inferential targets include the number of regimes, the changepoint locations, the segment-specific parameters, and uncertainty in all of these quantities. Bayesian formulations are particularly useful because they average over partitions, provide posterior probabilities for changepoints and segment counts, and propagate segmentation uncertainty to posterior prediction and signal estimation (Barry & Hartigan, 1992, 1993; Fearnhead, 2006; Hutter, 2007).

A central fact in Bayesian multiple-changepoint analysis is that exact posterior computation is possible when two conditions hold. First, the data contribution of a candidate segment can be integrated with respect to its segment-parameter prior. Second, the prior on an ordered partition factorizes into segmentwise or transitionwise terms. Product partition models express precisely this factorization (Barry & Hartigan, 1992, 1993). Exact recursions then sum over all admissible partitions without enumerating them explicitly (Fearnhead, 2006; Hutter, 2007). The statistical calculation for a segment and the combinatorial calculation over partitions are therefore distinct parts of the model.

This separation is especially consequential for exponential families. For a regular exponential-family sampling model, a proper conjugate prior updates through sufficient statistics and yields a segment marginal likelihood as a ratio of normalizing constants (Diaconis & Ylvisaker, 1979; Bernardo & Smith, 1994). Once the segment marginal likelihoods have been computed, the same partition recursions apply regardless of whether the observations are Gaussian, Poisson, Binomial, Negative Binomial, or another member of the class satisfying the stated regularity and integrability conditions. The generality of BayesBreak lies in this observation-family independence of the partition calculation, together with hierarchical extensions that preserve the required factorization.

BayesBreak develops that idea into a generalized hierarchical Bayesian segmentation method for irregular designs, multiple sequences, observed groups, and latent groups. The method retains the standard Bayesian multiple-changepoint formulation: segment parameters are integrated out, the resulting segment marginal likelihoods are combined with a prior on ordered partitions, and dynamic programming evaluates posterior sums or posterior maxima. Hierarchical structure enters through shared or group-specific partitions and through a finite latent-group model over segmentation templates. The method is exact when the segment integrals are exact and the partition prior has the stated factorization; when a segment integral is approximated numerically, the partition calculation is exact for the supplied approximate marginal-likelihood array, and the discrepancy from the target posterior is controlled only under an explicit segmentwise error bound.

1.1 Inferential quantities

For candidate segment counts $1 \leq k \leq k_{\max}$, the method distinguishes four inferential quantities:

1. the fixed-k marginal likelihood $p(y \mid k)$ and the segment-count posterior $p(k \mid y)$;
2. marginal posterior probabilities for changepoint events and ordered changepoint positions;
3. posterior moments of the piecewise-constant signal, including the Bayesian regression curve obtained by averaging over partitions; and
4. the joint maximum-a-posteriori partition, computed by max-sum dynamic programming and backtracking.

These quantities solve different decision problems. In particular, selecting the marginal mode of each ordered changepoint position does not generally recover the joint posterior mode and may fail to define a valid ordered partition.

1.2 Contributions

The paper makes the following methodological and computational contributions.

- ESTABLISHED **General exponential-family segment formulation.** We state an exposure-aware exponential-family model in which distributional descriptors, such as Poisson exposure or Binomial trial count, are separated from optional likelihood-power weights. A single conjugate calculation yields segment marginal likelihoods and posterior moments from cumulative sufficient statistics under the stated regularity conditions.
- ESTABLISHED **Posterior computation over ordered partitions.** Sum-product recursions compute fixed-count marginal likelihoods, the posterior distribution of the segment count, changepoint marginals, and posterior signal moments. A separate max-sum recursion with deterministic backtracking is proved to return the joint MAP partition. The worst-case time complexity is $\mathcal{O}(k_{\max} n^2)$ after segment quantities have been tabulated.
- ESTABLISHED **Irregular designs and multi-sequence hierarchies.** Design-dependent segment cohesions and candidate-boundary hazards are treated as distinct prior factors. For aligned conditionally independent sequences with a common partition, subject-specific segment marginal likelihoods multiply before the same partition recursion is applied. A finite-candidate consistency result is given under a positive average expected log-score gap.
- ESTABLISHED **Known-group and latent-group segmentation.** Known groups are handled by group-wise pooling. For unknown memberships, we define a finite latent-group template objective and derive a Jensen minorization, normalized auxiliary weights, and exact responsibility-weighted max-sum updates. The resulting algorithm is a minorization–maximization procedure for the stated objective, not a claim of full Bayesian averaging over latent partitions.
- CONDITIONAL **Nonconjugate segment models.** Laplace, variational, expectation-propagation, Pólya–Gamma, or quadrature calculations may supply approximate segment marginal likelihoods. We prove how a uniform error in reachable segment log marginal likelihoods propagates to partition marginal likelihoods and posterior odds; no approximation-specific convergence rate is asserted without additional assumptions.
- EXECUTED RESULT **Executed simulations and case studies.** We report the archived simulation and real-data outputs as executed. Comparator values are interpreted according to their actual reference sets, and computations using an incompatible coordinate axis or the wrong predictive family are excluded from the conclusions rather than reinterpreted as valid statistical evidence.

1.3 Scope

The central model is offline segmentation on a finite ordered candidate set, with piecewise-constant segment parameters and conditional independence given the partition and segment parameters. The exact algorithms require finite segment marginal likelihoods and a partition prior that factorizes over admissible segments or transitions. The method is intended for moderate sequence lengths for which an $n \times n$ segment table is feasible. Online filtering, nonordered spatial partitions, unrestricted dependence between adjacent segment parameters, and general within-segment dynamics require other models or enlarged state spaces.

1.4 Organization

Section 2 relates the method to product partition models, exact Bayesian changepoint recursions, exponential-family conjugacy, irregular-design priors, and multi-sequence models. Sections 3 and 4 define the segment models and posterior recursions. Section 5 develops irregular-design, shared-boundary, known-group, and latent-group extensions. Section 6 treats nonconjugate segment calculations and posterior prediction. Section 7 gives algorithms, complexity, and numerical checks. Sections 8 and 9 describe the executed experiments, followed by limitations and conclusions.

2 Statistical foundations and related work

2.1 Exponential families and conjugate segment models

The exponential-family representation separates a finite-dimensional sufficient statistic from the log normalizer of the sampling distribution. Diaconis and Ylvisaker characterize conjugate priors for natural exponential families and the hyperparameter conditions under which those priors are proper (Diaconis & Ylvisaker, 1979). In a changepoint model, the same calculation is applied separately to every candidate segment. BayesBreak uses this standard conjugate theory; its contribution is to formulate the resulting segment marginal likelihoods and posterior moment numerators so that they can be reused by the same partition recursions and by the hierarchical extensions developed below. Exactness requires the stated regularity, propriety, and finiteness conditions. The method does not assert closed-form integration for arbitrary nonconjugate parameterizations.

2.2 Product partition models and exact Bayesian recursions

Product partition models assign a partition probability proportional to a product of segment cohesions and combine this prior with segmentwise integrated likelihoods (Barry & Hartigan, 1992, 1993). This is the principal probabilistic lineage of the present work. Fearnhead developed exact recursions and posterior simulation for a broad class of Bayesian multiple-changepoint models with conditionally independent segments (Fearnhead, 2006); Hutter derived exact Bayesian regression for piecewise-constant functions, including posterior break probabilities and model-averaged regression curves (Hutter, 2007). Dependence between adjacent segments can sometimes be accommodated by enlarging the recursion state (Fearnhead & Liu, 2011). Bayesian Blocks provides a related local-fitness and dynamic-programming construction for astronomical time series (Scargle et al., 2013).

BayesBreak does not claim priority for product partition models, conjugate segment integration, exact changepoint recursions, posterior changepoint probabilities, or Bayesian piecewise-constant regression. It develops a generalized hierarchical model in which these established calculations are stated with common notation across exponential-family segment models, irregular-design priors, multiple sequences, observed groups, and a finite latent-group extension. It also keeps posterior marginalization and joint MAP optimization as separate recursions rather than treating one as an approximation to the other.

2.3 Frequentist segmentation and optimization

Dynamic programming also underlies frequentist segmentation with additive segment costs (Auger & Lawrence, 1989; Jackson et al., 2005). Penalized formulations and pruning lead to PELT and related pruned exact algorithms (Killick et al., 2012; Rigaill, 2015). Wild binary segmentation and SMUCE pursue different detection and uncertainty goals (Fryzlewicz, 2014; Frick et al., 2014). Total variation and the fused lasso replace discrete partition search with convex regularization (Rudin et al., 1992; Tibshirani et al., 2005); the group fused lasso encourages aligned changes across several sequences (Bleakley & Vert, 2011). Circular binary segmentation is a standard comparator for array-CGH data (Olshen et al., 2004). These methods are important computational and empirical references, but their estimands are penalized or testing-based rather than the posterior distribution over partitions considered here.

2.4 Irregular designs and design-dependent partition priors

Generalized product partition models and product partition models with covariates show how covariate information can alter partition probabilities through cohesion functions (Park & Dunson, 2010; Müller et al., 2011). BayesBreak specializes this idea to ordered observations. It distinguishes a segment-span cohesion from a candidate-boundary hazard because they encode different prior assumptions. Both factors can be included in the ordered-partition probability when they remain local to a segment or transition. Neither is treated as a likelihood weight.

2.5 Hierarchical and multi-sequence changepoint models

Hierarchical Bayesian changepoint analysis has a long history (Carlin et al., 1992). BASIC models the tendency of changepoints to co-occur across multiple sequences through a hierarchical empirical-Bayes prior and provides posterior sampling and optimization algorithms (Fan & Mackey, 2017). Joint random partition models provide a Bayesian nonparametric construction for dependent partitions across several ordered processes (Quinlan et al., 2024). BayesBreak studies a more restrictive setting for exact finite computation: aligned sequences are conditionally independent given a common partition, so their segment marginal likelihoods multiply. Known groups apply this calculation separately by group. The finite-candidate consistency theorem in Section 5 concerns this common-partition model and does not imply consistency for arbitrary hierarchical dependence.

2.6 Latent groups and common structural changes

Model-based clustering of time-dependent observations according to common structural changes is an active neighboring literature (Corradin et al., 2026). The BayesBreak latent-group construction is narrower than a full probability mixture over all partitions. It optimizes a finite objective of the form $\sum_s \log \sum_g \pi_g S_s(\tau_g)$, where $S_s(\tau_g) > 0$ is the specified subject–template score. A Jensen minorization yields auxiliary weights and responsibility-weighted max-sum updates. Because the S_s need not be normalized densities over the data space, the algorithm is described as minorization–maximization for a finite latent-group objective, not as maximum likelihood for an identifiable finite mixture (Dempster et al., 1977; Teicher, 1963).

2.7 Nonconjugate segment calculations

When a segment integral is unavailable in closed form, deterministic approximations such as Laplace approximation, the Jaakkola–Jordan variational bound, expectation propagation, Pólya–Gamma augmentation, or numerical quadrature may be used (Jaakkola & Jordan, 2000; Minka, 2001; Polson et al., 2013). The ordered-partition recursion is unchanged after an approximate segment marginal-likelihood table is supplied, but its output is then exact only for that table. Section 6 therefore proves a conditional error-propagation result: a uniform bound on reachable segment log marginal likelihoods yields bounds on partition marginal likelihoods and posterior odds. The theorem does not supply the segmentwise bound for a particular approximation routine.

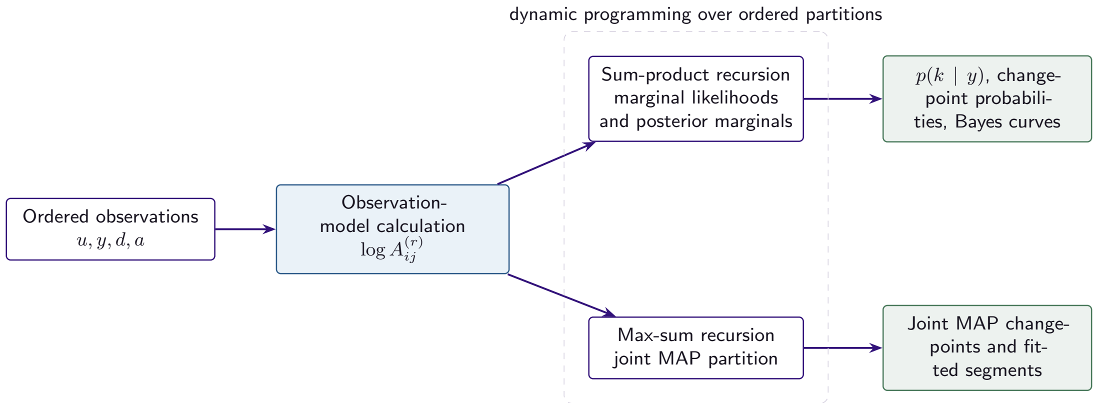


Figure 1. Statistical decomposition used by BayesBreak. A family-specific segment calculation supplies the integrated likelihood and optional posterior moment numerators. Local prior factors multiply the same triangular array. Sum-product and max-sum recursions over ordered partitions then produce posterior summaries and a joint MAP partition, respectively.

2.8 Position of the present work

The present contribution is a generalized hierarchical Bayesian segmentation model whose observation-family dependence is confined to segment marginal likelihoods and posterior moments. The paper derives the common posterior recursions, the separate MAP recursion, the irregular-design prior factors, exact shared-boundary pooling, the finite latent-group optimization, and the conditional approximation bounds in one notation. Its empirical claims concern the executed simulations and case studies reported here; it does not claim a universal complexity improvement or superiority over every established changepoint method.

3 Generalized segment models and integrated likelihoods

3.1 Ordered partitions and support

Let $y_{1:n}$ be observed on an ordered design. Candidate boundary coordinates are denoted $u_0 < u_1 < \cdots < u_n$; on a regular grid, $u_i = i$. A k-segment partition is a vector $t = (t_0, \ldots, t_k)$ with

$$0 = t_0 < t_1 < \cdots < t_k = n, \qquad I_q(t) = \{t_{q-1} + 1, \ldots, t_q\}.$$

The half-open notation $(i, j]$ denotes indices $i + 1, \ldots, j$. Physical block length is $\Delta_x(i, j) = u_j - u_i$ and is never inferred from an undefined observation x_0.

> *Assumption* 3.1 (Offline structural support and block independence). The candidate segment-count set $\{1, \ldots, k_{\max}\}$ is finite. Structural admissibility of a segment is fixed by the candidate grid, declared length restrictions, and prior support before the response values are observed. A structurally admissible segment has finite nonnegative local evidence; zero evidence removes its likelihood contribution but does not alter the partition-prior normalizer. Structurally excluded segments have local prior factor zero and log-score $-\infty$. Conditional on a partition, segment parameters are independent across segments. The conditional partition prior factorizes through local terms as in Assumption 3.4. For approximate segment routines, the same statements define an exact posterior for the supplied approximate score array; comparison to the intended nonconjugate model additionally requires Assumption 6.1.

3.2 Exponential-family segment likelihoods with exposure and power weights

Let d_r collect scientifically specified family information, such as Poisson exposure, Binomial trial count, or Gaussian precision. Let $a_r \geq 0$ be an optional power or reliability weight that multiplies the complete log contribution. The canonical local likelihood is

$$\begin{aligned} \log p(y_r \mid \theta, d_r)^{a_r} = a_r \log h(y_r, d_r) \\ + a_r S(y_r, d_r)^\top \eta(\theta) \\ - a_r E(d_r) A(\theta). \end{aligned} \tag{1}$$

The corresponding block summaries are

$$\begin{aligned} S_{ij} &= \sum_{r=i+1}^{j} a_r S(y_r, d_r), \\ W_{ij} &= \sum_{r=i+1}^{j} a_r E(d_r), \\ H_{ij} &= \sum_{r=i+1}^{j} a_r \log h(y_r, d_r). \end{aligned} \tag{2}$$

This notation prevents three distinct quantities from being collapsed into a single scalar "weight": design spacing affects the partition prior, d_r changes the observation law, and a_r raises the entire observation contribution to a power.

For conjugate hyperparameters (α, β), use the Diaconis–Ylvisaker normalizer convention

$$\begin{aligned} p(\theta \mid \alpha, \beta) &= \exp\{\eta(\theta)^\top \alpha - \beta A(\theta) - \log Z(\alpha, \beta)\}, \\ Z(\alpha, \beta) &= \int e^{\eta(\theta)^\top \alpha - \beta A(\theta)} \, \mathrm{d}\theta. \end{aligned} \tag{3}$$

Fix a scientifically declared reference descriptor $d_\star$ when the conditional observation mean depends on exposure or trial structure, and write $m_\star(\theta) = m(\theta; d_\star)$ for the segment functional reported by the analysis (for example, a Poisson rate at unit exposure rather than an exposure-specific count mean). Define

$$\begin{aligned} A_{ij}^{(0)} &= p(y_{i+1:j} \mid d_{i+1:j}, a_{i+1:j}), \\ A_{ij}^{(r)} &= A_{ij}^{(0)} \, \mathbb{E}[m_\star(\theta)^r \mid y_{i+1:j}]. \end{aligned}$$

Theorem 3.2 (Exposure-aware conjugate block integration)**.** *If $Z(\alpha_0 + S_{ij}, \beta_0 + W_{ij}) < \infty$, then*

$$A_{ij}^{(0)} = e^{H_{ij}} \frac{Z(\alpha_0 + S_{ij}, \beta_0 + W_{ij})}{Z(\alpha_0, \beta_0)}. \tag{4}$$

The block posterior is conjugate with hyperparameters $(\alpha_0 + S_{ij}, \beta_0 + W_{ij})$. Whenever its rth moment exists,

$$A_{ij}^{(r)} = A_{ij}^{(0)} \mathbb{E}[m_\star(\theta)^r \mid \alpha_0 + S_{ij}, \beta_0 + W_{ij}]. \tag{5}$$

Proof. Multiplying the powered likelihood terms in Equation (1) gives $e^{H_{ij}} \exp\{\eta(\theta)^\top S_{ij} - W_{ij} A(\theta)\}$. Multiplication by the prior in Equation (3) produces the same conjugate kernel with updated hyperparameters $(\alpha_0 + S_{ij}, \beta_0 + W_{ij})$. Integrating the kernel yields the ratio of normalizers in Equation (4); division by this evidence gives the posterior. Inserting $m_\star(\theta)^r$ into the integral then factors the result into the evidence times the posterior moment, proving Equation (5). □

Remark 3.3 (Scope of exactness). Theorem 3.2 is exact for the displayed powered, exposure-aware model.

Table 1. Supported local block constructions. Closed-form rows instantiate Theorem 3.2; the Beta-valued response uses deterministic one-dimensional quadrature and is therefore exact only up to its numerical tolerance.

Family	Descriptor d_r	Segment parameter	Integrated block construction	Status
Gaussian, known variance	precision or inverse variance	mean μ	Gaussian–Gaussian normalizer and posterior moments	closed form
Poisson	exposure e_r	rate λ	Gamma–Poisson normalizer; $E(d_r) = e_r$	closed form
Binomial	trial count m_r	probability p	Beta–Binomial normalizer; $E(d_r) = m_r$	closed form
Negative Binomial	fixed dispersion	success parameter	Beta-conjugate distributional parameterization	closed form
Beta-valued response	known precision ϕ_r	mean $\mu \in (0,1)$	one-dimensional integral over a transformed mean	quadrature
Nonconjugate GLM	link-specific inputs	regression parameter	Laplace, variational, EP, PG, or quadrature block score	approximate unless certified

It does not permit a Binomial trial count to be treated as a likelihood power, an irregular coordinate gap to be inserted as exposure, or a numerical integral to be described as conjugate closed form.

3.3 Local prior factors

Assumption 3.4 (Factorized ordered-partition prior). For fixed k and boundary coordinates u, the prior on t is

$$p(t \mid k, u) = C_k(u)^{-1} \prod_{q=1}^{k} c_u(t_{q-1}, t_q) \prod_{q=1}^{k-1} h_u(t_q), \tag{6}$$

where $c_u(i,j) \geq 0$ is a complete-segment cohesion and $h_u(j) \geq 0$ is an interior-boundary hazard. Define $\gamma_u(i,j) = c_u(i,j) h_u(j)^{\mathbb{1}\{j<n\}}$ and $\widetilde{A}_{ij}^{(r)} = A_{ij}^{(r)} \gamma_u(i,j)$.

The normalizer $C_k(u)$ is itself a partition sum and can be computed by the same dynamic program after replacing all likelihood evidences by one. This ensures that fixed-k evidence remains properly normalized for nonuniform local priors.

3.4 Segment marginal-likelihood interface

The partition recursions require only a triangular array of segment log marginal likelihoods and, when posterior signal moments are requested, corresponding segment moment summaries. A family-specific implementation returns the segment log marginal likelihood, posterior hyperparameters or moments, a validity flag, and any numerical diagnostic needed to interpret an approximate value. Prefix summaries make Gaussian, Poisson, and Binomial segment calculations constant time after $\mathcal{O}(n)$ preprocessing. The recursion over partitions is therefore unchanged when the observation family changes.

Required segment quantities

For every candidate segment $(i,j]$, the family-specific calculation returns $\left(\log A_{ij}^{(0)}, \mathcal{M}_{ij}, \texttt{status}_{ij}, \texttt{diagnostics}_{ij}\right)$, where $\mathcal{M}_{ij}$ contains the moments or posterior

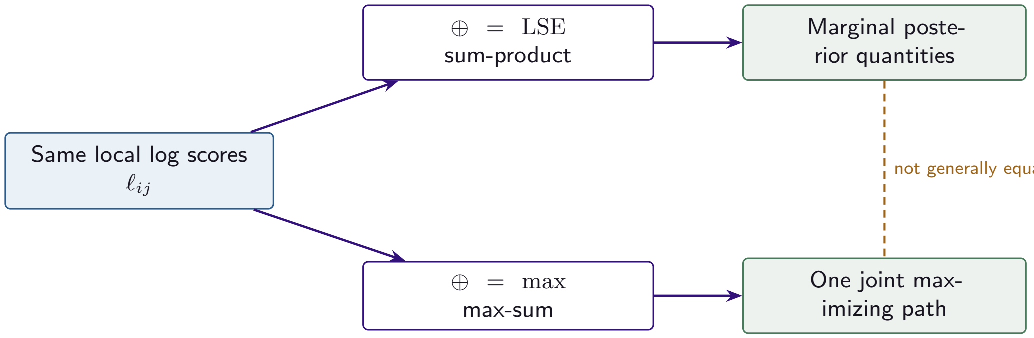


Figure 2. The same local scores support two algebraically related but inferentially different routes. Sum-product integrates over paths; max-sum selects one globally maximizing path. Coordinatewise modes of marginal boundary distributions need not form that path.

> state required downstream. A segment outside the declared support receives log marginal likelihood $-\infty$, not an arbitrary finite penalty. Numerical approximations additionally report convergence information and a reference-error estimate whenever one is available.

4 Posterior computation over ordered partitions

Once the prior-adjusted block array $\widetilde{A}^{(r)}_{ij} = A^{(r)}_{ij}\gamma_u(i,j)$ has been constructed, the observation family no longer enters the partition recursions. This section gives the two distinct inference routes used by BAYESBREAK: sum-product for posterior quantities and max-sum for a joint maximizing partition.

4.1 Forward and backward partition sums

For $k \in \{0,\dots,k_{\max}\}$ and $j \in \{0,\dots,n\}$, let $L_{k,j}$ be the integrated score of all k-block partitions of the prefix $(0,j]$. Let $R_{k,i}$ be the corresponding score for all k-block partitions of the suffix $(i,n]$. Set

$$
\begin{aligned}
L_{0,0} &= 1, \qquad L_{0,j} = 0 \quad (j>0),\\
R_{0,n} &= 1, \qquad R_{0,i} = 0 \quad (i<n).
\end{aligned}
$$

The sum-product recursions are

$$
L_{k+1,j} = \sum_{h=k}^{j-1} L_{k,h}\widetilde{A}^{(0)}_{h,j}, \tag{7}
$$

$$
R_{k+1,i} = \sum_{h=i+1}^{n-k} \widetilde{A}^{(0)}_{i,h} R_{k,h}. \tag{8}
$$

In implementation, Equations (7)–(8) are evaluated in log space using `logsumexp`.

> **Theorem 4.1** (Partition-sum identities and boundary marginals)**.** *Under Assumptions 3.1 and 3.4, for every admissible k,*
>
> $$L_{k,n} = R_{k,0} = \sum_{0=t_0<\cdots<t_k=n} \prod_{q=1}^{k} \widetilde{A}^{(0)}_{t_{q-1},t_q}. \tag{9}$$

Moreover, for $1 \le p \le k-1$ and admissible h,

$$p(t_p = h \mid y, k) = \frac{L_{p,h} R_{k-p,h}}{L_{k,n}},$$
$$p(b_h = 1 \mid y, k) = \sum_{p=1}^{k-1} p(t_p = h \mid y, k). \tag{10}$$

Proof. Induction on k in Equation (7) shows that $L_{k,j}$ sums the product of local scores over every ordered k-block path ending at j. Backward induction gives the same statement for $R_{k,i}$ on suffixes, proving Equation (9). A partition satisfying $t_p = h$ has a unique decomposition into a p-block prefix ending at h and a $(k-p)$-block suffix beginning at h. Their integrated scores multiply, and division by the total partition sum yields the first identity in Equation (10). Strict ordering permits at most one boundary label p at any fixed h within a partition, so summing over p gives the boundary-event probability. □

Let $p(k)$ be a segment-count prior. Because $C_k(u)$ normalizes Equation (6),

$$p(k \mid y) \propto p(k) \frac{L_{k,n}}{C_k(u)}, \qquad k = 1, \ldots, k_{\max}. \tag{11}$$

The posterior may be reported in full; choosing $\widehat{k} = \arg\max_k p(k \mid y)$ is a separate decision rule. Conditional boundary distributions in Equation (10) and boundary-event probabilities can then be computed at fixed k or averaged over $p(k \mid y)$.

4.2 Posterior moments and Bayes curves

For $r \in \{0, 1, 2\}$, define a block-covering contribution at fixed k,

$$F_{ij}^{(r)}(k) = \frac{1}{L_{k,n}} \sum_{m=1}^{k} L_{m-1,i} \widetilde{A}_{ij}^{(r)} R_{k-m,j}. \tag{12}$$

When the target at coordinate s is the segment-level quantity $m_\star(\theta)$, its conditional posterior moments are

$$\mathbb{E}[m(\theta_s)^r \mid y, k] = \sum_{i<s\le j} F_{ij}^{(r)}(k). \tag{13}$$

The posterior mean Bayes curve uses $r = 1$; the $r = 2$ result gives pointwise posterior variance after subtracting the squared mean.

Proposition 4.2 (Block-covering decomposition)**.** *If every admissible partition assigns each coordinate to exactly one block and $A_{ij}^{(r)}$ is the integrated rth moment numerator for that block, then Equation (13) is the exact fixed-k posterior moment of the segment-level target at s.*

Proof. Every partition contains one and only one block $(i, j]$ satisfying $i < s \le j$. For a specified block position m, the score of all compatible prefixes is $L_{m-1,i}$, the integrated target contribution on $(i, j]$ is $\widetilde{A}_{ij}^{(r)}$, and the score of all compatible suffixes is $R_{k-m,j}$. Summing over m, then over all blocks covering s, enumerates each k-block partition exactly once with its integrated target numerator. Division by $L_{k,n}$ normalizes the posterior. □

4.3 Joint MAP segmentation

Define local log scores $\ell_{ij} = \log \widetilde{A}_{ij}^{(0)}$ and max-sum messages

$$M_{k,j} = \max_{h \in \{k-1,\ldots,j-1\}} \{M_{k-1,h} + \ell_{h,j}\}, \quad M_{0,0} = 0, \tag{14}$$

with $M_{0,j} = -\infty$ for $j > 0$. Store a deterministic maximizing predecessor $h^\star(k, j)$. Starting at (k, n) and following these predecessors recovers a complete boundary vector.

Theorem 4.3 (Correctness of max-sum backtracking)**.** *For fixed k, the boundary vector returned by Equation (14) and its stored predecessors belongs to*

$$\arg \max_{0=t_0<\cdots<t_k=n} \sum_{q=1}^{k} \ell_{t_{q-1},t_q}. \tag{15}$$

Consequently, it is a joint MAP segmentation conditional on k. Ties are resolved only by the declared predecessor convention.

Proof. For $k = 1$, $M_{1,j} = \ell_{0,j}$ is the only one-block path score. Suppose $M_{k-1,h}$ is the optimal score among all $(k-1)$-block paths ending at h. Every k-block path ending at j has a unique penultimate endpoint h and total score equal to its prefix score plus $\ell_{h,j}$. Maximizing over h therefore produces the optimal k-block score. Induction proves optimality at (k, n). Each stored predecessor attains the corresponding maximum, so backtracking constructs a path attaining that terminal score. □

Marginal modes are not a joint MAP path

The vector formed from $\arg\max_h p(t_p = h \mid y, k)$ for each p optimizes separate marginal decisions. Its entries can be incompatible or can have lower joint posterior probability than another partition. BayesBreak therefore reports marginal modes as diagnostics and reserves "joint MAP" for Theorem 4.3.

4.4 Complexity and numerical invariants

If cumulative sufficient statistics make each local conjugate score constant time, there are $\Theta(n^2)$ blocks to precompute. Sum-product and max-sum each inspect $\Theta(k_{\max} n^2)$ predecessor transitions. Values and backpointers require $\Theta(k_{\max} n)$ working storage, while retaining all block-cover terms can require $\Theta(k_{\max} n^2)$ storage.

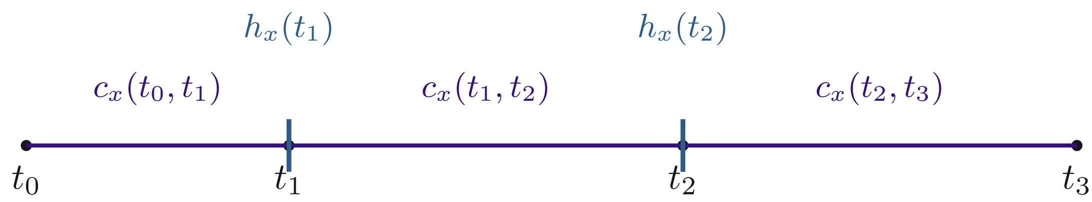
Proposition 4.4 (Complexity)**.** *When cumulative sufficient statistics make each conjugate segment calculation constant time, segment precomputation costs $\Theta(n^2)$ time and storage; either partition recursion costs $\Theta(k_{\max} n^2)$ time. For a nonconjugate block routine with per-block cost T_{block}, local precomputation costs $\Theta(n^2 T_{\text{block}})$ and the global recursions are unchanged.*

Proof. There are $n(n + 1)/2 = \Theta(n^2)$ admissible intervals. For each k and endpoint j, a recursion scans at most j predecessors, so $\sum_{k=1}^{k_{\max}} \sum_{j=k}^{n} \Theta(j) = \Theta(k_{\max} n^2)$ when all blocks are admissible. Replacing constant-time local evaluation by cost T_{block} multiplies only the block-construction term. □

Four numerical identities are checked in every implementation: the normalized segment-count posterior sums to one; forward and backward total evidence agree; conditional boundary-event probabilities sum to $k - 1$; and the backtracked path score equals the terminal max-sum score within numerical tolerance. These checks test algebraic implementation rather than empirical model quality.

5 Irregular designs and hierarchical extensions

The ordered-partition recursions remain valid when prior factors or multiple-sequence likelihoods can be expressed segment by segment. This section gives the corresponding models for irregular coordinates, shared changepoints, observed groups, and latent groups.



$p(t \mid k, x) \propto \prod_q c_x(t_{q-1}, t_q) \prod_{q<k} h_x(t_q)$; the terminal endpoint receives no boundary-hazard factor.

Figure 3. Design-dependent partition prior. Segment cohesion and interior-boundary hazard represent different prior assumptions. The terminal endpoint receives no boundary-hazard factor.

5.1 Irregular designs: segment cohesion and boundary hazard

Irregular coordinates can enter the partition prior in two statistically distinct ways. A complete-segment cohesion $c_u(i, j)$ assigns prior mass according to the physical span of $(i, j]$; an interior-boundary hazard $h_u(j)$ assigns prior mass to a candidate split at u_j. They can appear together in Equation (6), but one does not imply the other.

> **Theorem 5.1** (Local prior factors preserve exact inference)**.** *Replacing* $A_{ij}^{(r)}$ *by* $\widetilde{A}_{ij}^{(r)} = A_{ij}^{(r)} c_u(i, j) h_u(j)^{\mathbb{1}\{j<n\}}$ *causes the sum-product recursion to enumerate exactly the unnormalized joint weights in Equation* (6)*. Dividing the terminal partition sum by* $C_k(u)$ *yields the proper fixed-*k *marginal likelihood.*

Proof. For any partition t, the product of modified segment quantities is

$$\prod_{q=1}^{k} A_{t_{q-1},t_q}^{(r_q)} \prod_{q=1}^{k} c_u(t_{q-1}, t_q) \prod_{q=1}^{k-1} h_u(t_q),$$

because the terminal indicator removes $h_u(n)$. This is the likelihood or moment product multiplied by the unnormalized partition prior. The recursion visits each ordered boundary vector once; normalization by $C_k(u)$ completes the proof. □

A boundary-process interpretation applies only to the hazard factor. Let $\Lambda_j = \int_{u_{j-1}}^{u_j} \lambda(v)\,\mathrm{d}v$ for an inhomogeneous Poisson process. Interval occupancies are independent Bernoulli variables with probability $1 - e^{-\Lambda_j}$. Conditional on exactly m occupied candidate intervals, the subset probability is proportional to the product of Bernoulli odds, so the compatible local factor is

$$h_u(j) \propto \frac{1 - e^{-\Lambda_j}}{e^{-\Lambda_j}} = e^{\Lambda_j} - 1. \tag{16}$$

Constant intensity and equal-width intervals recover an index-uniform fixed-count boundary prior when $c_u \equiv 1$. Without conditioning on the number of occupied intervals, both occupied and unoccupied Bernoulli factors are part of the prior. Equation (16) does not justify multiplying a segment likelihood by its physical length; a duration-dependent cohesion is a separate prior choice.

5.2 Shared changepoints across multiple sequences

Suppose S aligned sequences share a boundary vector while retaining sequence-specific segment parameters and priors. Conditional independence gives a pooled segment marginal likelihood.

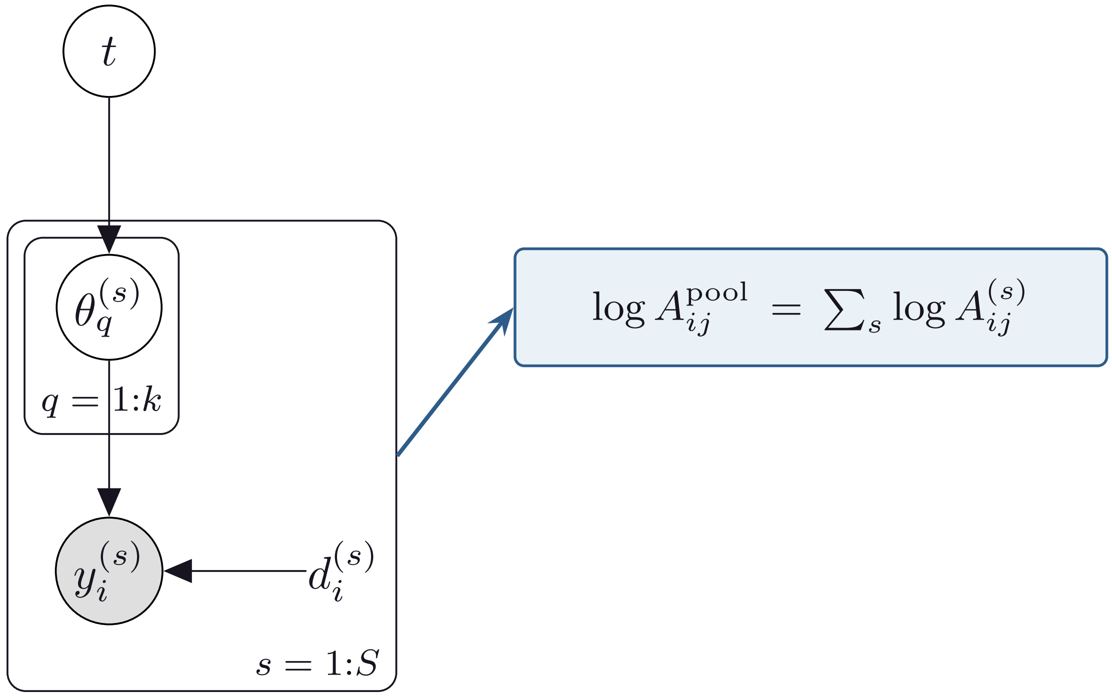


Figure 4. Hierarchical common-changepoint model with sequence-specific segment parameters. Conditional independence makes the sequence-specific segment marginal likelihoods multiply.

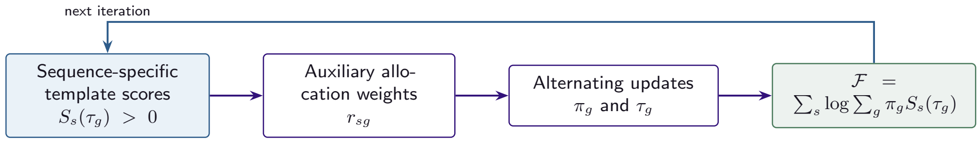
> *Assumption* 5.2 (Common candidate set). The sequences are aligned to a common ordered candidate set. Every candidate partition considered by the common-changepoint model is admissible for every sequence after the declared missing-data or exposure convention. Conditional on the common partition, sequences and their segment parameters are independent, and all sequence-specific segment marginal likelihoods are finite and positive on the common support.

> **Theorem 5.3** (Exact pooling under a common partition)**.** *Under Assumption 5.2, the pooled segment marginal likelihood is*
>
> $$A_{ij}^{(0),\text{pool}} = \prod_{s=1}^{S} A_{ij}^{(0,s)}, \qquad \ell_{ij}^{\text{pool}} = \sum_{s=1}^{S} \ell_{ij}^{(s)}. \tag{17}$$
>
> *Using this array in Section 4 gives the exact posterior distribution over the common partition. Conditional sequence-specific parameter posteriors are recovered from the original sequence-specific conjugate updates.*

Proof. For a fixed candidate segment, sequence-specific parameters integrate independently, producing $A_{ij}^{(0,s)}$ for sequence s. Conditional independence makes the joint integrated segment contribution their product. The partition recursion depends only on the resulting triangular array, so its path sum is the exact posterior normalizer for the common-partition model. The sequence-specific posterior states remain available conditional on any selected or averaged partition. □

Known groups are handled by applying Equation (17) separately within each observed group. The following asymptotic statement is restricted to a finite candidate set and requires information to accumulate across sequences.

> **Theorem 5.4** (Finite-candidate consistency for a common partition)**.** *Let $\mathcal{T}$ be finite and contain a data-generating candidate $t^\star$. For sequence s, let $M_s(t) > 0$ be the integrated score used by the common-partition model. Assume independent sequences, $p(t^\star) > 0$, and for each $t \neq t^\star$:*
>
> 1. $\mathbb{E}|\log M_s(t^\star) - \log M_s(t)| < \infty$*;*



The updates maximize a Jensen minorizer of the stated finite latent-group criterion. The criterion is not interpreted as a normalized finite-mixture likelihood.

Figure 5. Minorization–maximization for the finite latent-group criterion. The auxiliary weights are normalized across templates, and the template step is solved by responsibility-weighted max-sum dynamic programming.

> 2. *a strong law holds for the centered log-score differences; and*
> 3. $\liminf_{S\to\infty} S^{-1}\sum_{s=1}^{S} \mathbb{E}[\log M_s(t^\star) - \log M_s(t)] = \delta_t > 0.$
>
> *Then* $p(t^\star \mid y^{(1:S)}) \to 1$ *almost surely, and every joint MAP segmentation is eventually* $t^\star$ *almost surely.*

Proof. For $t \neq t^\star$, define $D_s(t) = \log M_s(t^\star) - \log M_s(t)$. The posterior ratio is

$$\frac{p(t \mid y^{(1:S)})}{p(t^\star \mid y^{(1:S)})} = \frac{p(t)}{p(t^\star)} \exp\left\{-\sum_{s=1}^{S} D_s(t)\right\}.$$

The strong law and positive limiting average gap make the normalized sum have positive liminf δ_t almost surely, so the ratio converges to zero exponentially on the S scale. Finiteness of $\mathcal{T}$ permits summing over all incorrect candidates. Posterior normalization then gives concentration at $t^\star$, and every incorrect posterior ratio is eventually below one, proving the MAP statement. □

One fixed informative sequence is insufficient for this theorem because its contribution vanishes in the average as S increases. The assumption requires an accumulating average expected log-score gap.

5.3 Finite latent-group template model

For sequence s and deterministic segmentation template $\tau = (k, t)$, the archived implementation uses

$$S_s(\tau) = \frac{p(k)}{C_k} \prod_{q=1}^{k} A^{(0,s)}_{t_{q-1},t_q} \gamma_u(t_{q-1}, t_q). \tag{18}$$

These subject-template scores are not required to integrate to one over the data space. We therefore define the finite criterion

$$\begin{aligned} \mathcal{F}(\pi,\tau) &= \sum_{s=1}^{S} \log\left\{\sum_{g=1}^{G} \pi_g S_s(\tau_g)\right\}, \\ &\pi_g \geq 0, \qquad \sum_g \pi_g = 1. \end{aligned} \tag{19}$$

This is a latent-group clustering criterion. It is not interpreted as a finite-mixture likelihood unless each $S_s(\tau_g)$ is a normalized sampling density.

For row-stochastic auxiliary weights r_{sg}, Jensen's inequality gives

$$\begin{aligned} &\mathcal{F}(\pi,\tau) \geq \mathcal{Q}(r,\pi,\tau), \\ &\mathcal{Q}(r,\pi,\tau) = \sum_{s,g} r_{sg}\left[\log \pi_g + \log S_s(\tau_g) - \log r_{sg}\right], \end{aligned} \tag{20}$$

with equality at

$$r_{sg} = \frac{\pi_g S_s(\tau_g)}{\sum_h \pi_h S_s(\tau_h)}. \tag{21}$$

The mixing-weight update is $\pi_g = S^{-1} \sum_s r_{sg}$. Let $n_g = \sum_s r_{sg}$. For a candidate segment count k, substitution of Equation (18) into the minorizer gives the count offset $n_g\{\log p(k) - \log C_k\}$ and the max-sum local score

$$\ell_{ij}^{(g)} = \sum_{s=1}^{S} r_{sg} \log A_{ij}^{(0,s)} + n_g \log \gamma_u(i,j). \tag{22}$$

The template update solves the fixed-k max-sum problem and then compares the offset-adjusted terminal values across the allowed segment counts.

Theorem 5.5 (Monotonicity of exact latent-group updates)**.** *An iteration that sets r by Equation* (21)*, maximizes $\mathcal{Q}$ over π, and exactly maximizes every responsibility-weighted template criterion using Equation* (22) *and its segment-count offset over the declared finite candidate set cannot decrease $\mathcal{F}$.*

Proof. At the old parameters, Equation (21) makes the lower bound tight. Exact coordinate maximization cannot decrease $\mathcal{Q}$. Jensen's inequality guarantees that the new value of $\mathcal{F}$ is at least the new value of $\mathcal{Q}$. Chaining these inequalities gives $\mathcal{F}_{\text{new}} \geq \mathcal{F}_{\text{old}}$. □

Template-label permutations, coincident templates, and an oversized G can make the optimum non-unique. No finite-mixture identifiability or generative-model consistency claim is made. Review of the archived implementation also found an exit path that could return the previous rather than the final objective; one of three diagnostic restarts differed by 0.003137360860080. The archived simulation remains an executed computation, but reliable restart selection requires the final objective value from every run.

6 Approximate segment integration, posterior prediction, and evaluation

For a nonconjugate segment model, BayesBreak requires a numerical approximation to the segment marginal likelihood. Posterior prediction requires the posterior predictive distribution associated with the fitted observation model. This section states the error condition under which segmentwise approximation error can be propagated through the partition recursions and then defines the prediction and boundary-evaluation rules used in the empirical analysis.

6.1 Propagation of segment marginal-likelihood error

Let $\ell_{ij} = \log A_{ij}^{(0)}$ be a reference log segment marginal likelihood and let $\widehat{\ell}_{ij}$ be the value returned by a numerical method such as Laplace approximation, the Jaakkola–Jordan bound, expectation propagation, Pólya–Gamma mean-field approximation, or adaptive quadrature.

Assumption 6.1 (Uniform error bound over admissible segments)*.* There exists $\varepsilon \geq 0$ such that

$$|\widehat{\ell}_{ij} - \ell_{ij}| \leq \varepsilon \tag{23}$$

for every segment $(i,j]$ that can occur in a partition with $k \leq k_{\max}$. The numerical analysis must identify the reference value and, when applicable, separate optimization residual, truncation or tail error, quadrature error, and discrepancy from the reference calculation.

Theorem 6.2 (Stability of marginal likelihoods and posterior odds)**.** *Under Assumption 6.1, every*

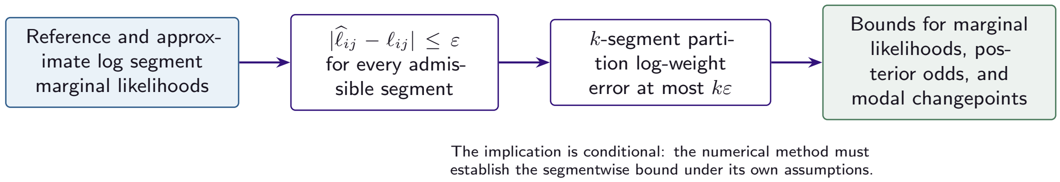


Figure 6. Propagation of a uniform log marginal-likelihood error through the partition recursions. The global bounds require an independently established segmentwise error bound; they do not provide such a bound for a numerical approximation.

k-segment forward or backward quantity satisfies

$$\begin{aligned} e^{-k\varepsilon} L_{k,j} \le \widehat{L}_{k,j} \le e^{k\varepsilon} L_{k,j}, \\ e^{-k\varepsilon} R_{k,i} \le \widehat{R}_{k,i} \le e^{k\varepsilon} R_{k,i}. \end{aligned} \tag{24}$$

Consequently, for $k, k' \le k_{\max}$,

$$\left| \log \frac{\widehat{p}(k \mid y)}{\widehat{p}(k' \mid y)} - \log \frac{p(k \mid y)}{p(k' \mid y)} \right| \le (k + k')\varepsilon, \tag{25}$$

and, for two locations h, h' of the same ordered boundary t_p under fixed k,

$$\left| \log \frac{\widehat{p}(t_p = h \mid y, k)}{\widehat{p}(t_p = h' \mid y, k)} - \log \frac{p(t_p = h \mid y, k)}{p(t_p = h' \mid y, k)} \right| \le 2k\varepsilon. \tag{26}$$

Proof. Each term in a k-segment partition sum is a product of exactly k segment marginal likelihoods. Equation (23) therefore changes every path weight by a factor in $[e^{-k\varepsilon}, e^{k\varepsilon}]$. Summation over positive path weights preserves this multiplicative bound, which proves Equation (24). Posterior odds for two segment counts divide terminal sums whose multiplicative errors are bounded by $e^{\pm k\varepsilon}$ and $e^{\pm k'\varepsilon}$, giving Equation (25). The unnormalized probability of an ordered boundary at h is $L_{p,h} R_{k-p,h}$ and therefore contains k segment factors in total. Comparing two boundary locations doubles the largest possible log perturbation and gives Equation (26). □

Corollary 6.3 (Consequences for probabilities and modal locations)**.** *If finite unnormalized weights satisfy $|\log \widehat{w}_a - \log w_a| \le \eta$, then their normalized probabilities satisfy $e^{-2\eta} \le \widehat{p}_a / p_a \le e^{2\eta}$ and $\|\widehat{p} - p\|_{\mathrm{TV}} \le \min\{1, e^{2\eta} - 1\}$. Moreover, a fixed-$k$ modal boundary separated from every alternative by exact log-odds margin Δ is unchanged whenever $\varepsilon < \Delta/(2k)$.*

Proof. Write $\widehat{w}_a = w_a e^{\delta_a}$ with $|\delta_a| \le \eta$. The normalizing multiplier $\sum_b p_b e^{\delta_b}$ lies in $[e^{-\eta}, e^{\eta}]$, which gives the probability-ratio bounds; summing positive probability differences yields the exponential bound, while total variation is at most one. Equation (26) perturbs each best-versus-alternative log-odds difference by at most $2k\varepsilon$. The ordering is therefore unchanged when $2k\varepsilon < \Delta$. □

Proof obligation 6.4 (Error control for individual numerical methods). This paper does not establish a uniform convergence rate for Laplace approximation, the Jaakkola–Jordan bound, expectation propagation, Pólya–Gamma mean-field approximation, or adaptive quadrature over all admissible segments. To apply Theorem 6.2, a nonconjugate analysis must establish Assumption 6.1 under conditions appropriate to the observation family, parameter domain, segment support, initialization, convergence event, and tail behavior. Without that step, the dynamic program is exact for the supplied approximate score array, but the theorem does not compare the resulting posterior with the posterior under the intended segment model.

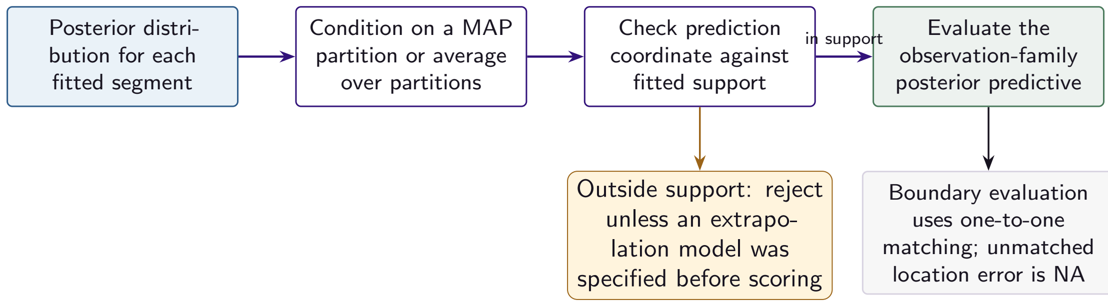


Figure 7. Posterior prediction and evaluation. Observation-family dispatch and coordinate-support checks precede proper scoring; a location error is undefined when no boundary is matched.

6.2 Family-specific posterior prediction

A fitted model records the observation family, the segment posterior state, and the coordinate range used for fitting. Prediction must use the posterior predictive distribution of that observation family. Coordinates outside the fitted range are rejected by default; an extrapolation rule is permitted only when it is stated as part of the predictive model before evaluation.

Consider a fitted segment B with posterior conjugate hyperparameters (α_B, β_B). Let $(S_B^{\mathrm{new}}, W_B^{\mathrm{new}}, H_B^{\mathrm{new}})$ denote the exposure-aware sufficient summaries of new observations assigned to B.

Proposition 6.5 (Posterior-predictive likelihood for a fixed segment)**.** *Under the exponential-family formulation of Section 3,*

$$p(y_B^{\mathrm{new}} \mid y_B) = e^{H_B^{\mathrm{new}}} \frac{Z(\alpha_B + S_B^{\mathrm{new}}, \beta_B + W_B^{\mathrm{new}})}{Z(\alpha_B, \beta_B)}. \tag{27}$$

Proof. Use the posterior distribution for the segment parameter after observing y_B as the prior distribution for the new observations. Multiplication by the new exposure-aware likelihood updates (α_B, β_B) by $(S_B^{\mathrm{new}}, W_B^{\mathrm{new}})$. Integration over the segment parameter gives the ratio of conjugate normalizing constants and the base-measure factor in Equation (27). □

Prediction may condition on an exported joint-MAP partition or average over partitions and segment counts. The first calculation is conditional on a selected partition; the second integrates segmentation uncertainty. In either case, assignment of prediction coordinates to segments is part of the statistical model. Assigning every out-of-range coordinate to the final segment without an explicit extrapolation assumption is not a valid posterior-predictive calculation.

6.3 Boundary metrics

Definition 6.6 (One-to-one tolerance matching)**.** For estimated boundaries $\widehat{\mathcal{B}}$, reference boundaries $\mathcal{B}^\star$, and tolerance τ, form the bipartite graph containing an edge $(\widehat{b}, b^\star)$ whenever $|\widehat{b} - b^\star| \le \tau$. Choose a matching of maximum cardinality; among maximum-cardinality matchings, choose one of minimum total absolute distance under a deterministic tie rule. Precision, recall, and F_1 use the resulting number of matches. No boundary is used more than once.

Definition 6.7 (Matched location error)**.** For the matching $\mathcal{M}_\tau$,

$$\mathrm{MAE}_\tau = |\mathcal{M}_\tau|^{-1} \sum_{(\widehat{b}, b^\star) \in \mathcal{M}_\tau} |\widehat{b} - b^\star|$$

when $|\mathcal{M}_\tau| > 0$. When no pair is matched, the value is reported as NA, not as zero.

Algorithm 1: BayesBreak offline fit

Input: Ordered observations $\{(u_r, y_r, d_r, a_r)\}_{r=1}^{n}$; observation family and hyperparameters; $k_{\max}$; segment-count prior $p(k)$; segment-cohesion and boundary-hazard factors c_u, h_u; numerical tolerances.

Output: Posterior distribution over k; boundary-event and ordered-boundary marginals; joint MAP partitions; segment posterior states; Bayes curves; numerical diagnostics and result metadata.

Validate order, observation-family support, descriptors, power weights, and candidate-segment restrictions;
Construct cumulative sufficient statistics when available;
for *each admissible segment* $(i, j]$ **do**
    compute $\log A_{ij}^{(0)}$ and the required moments or posterior state;
    record validity, convergence, and approximation diagnostics;
    assign score $-\infty$ to an inadmissible segment rather than substituting a finite value from another model;
Compute $\log \gamma_u(i,j) = \log c_u(i,j) + \mathbb{1}\{j < n\} \log h_u(j)$ once for each segment;
Run log-space forward and backward recursions and normalize Equation (11);
Compute ordered-boundary marginals, boundary-event probabilities, and Bayes-curve moments;
Run max-sum recursion and deterministic backtracking for every retained k or for the selected $\widehat{k}$;
Check forward–backward equality, posterior normalization, boundary-event mass, and backtrack-score equality;
Export the observation-family identifier, fitted coordinate range, result metadata, and numerical diagnostics;

Definition 6.8 (Admissible held-out predictive score)**.** A held-out log score is used as predictive evidence only when the held-out observations were excluded from fitting, the posterior predictive distribution of the fitted observation family was used, and every prediction coordinate lay in the fitted support or followed a pre-specified extrapolation model. A violation of any condition invalidates the number for that predictive interpretation without changing the fact that the computation was executed.

Excluded methylation predictive computation

The archived methylation value -387.50040013308154 for $m = 381$ is a real executed output, but it is not a Beta-observation held-out log predictive density. The implementation used a Gaussian predictive routine and assigned coordinates beyond the fitted range to the last segment. The value remains recorded as `RES-BB-RD-007Q` and is excluded from predictive conclusions.

7 Algorithms, computational complexity, and numerical implementation

The implementation separates calculations specific to an observation model from calculations over ordered partitions. Observation-model routines compute sufficient statistics, segment marginal likelihoods, posterior moments, numerical error information, and posterior-predictive quantities. The partition routines combine prior-adjusted segment scores, evaluate sum-product recursions, recover joint MAP partitions by max-sum recursion and backtracking, pool aligned sequences, and extract posterior summaries.

7.1 Inputs and returned quantities

The fit input separates four objects that have different statistical meanings: ordered coordinate u_r, response y_r, observation descriptor d_r, and optional likelihood power a_r. Descriptors such as exposure or trial count are interpreted by the corresponding observation model. Irregular geometry enters the prior through u and

Table 2. Principal returned quantities and their inferential interpretation.

Quantity	Calculation	Interpretation	Numerical checks
$\log A_{ij}^{(0)}$	observation-model segment calculation	log integrated likelihood or declared numerical approximation	support, finiteness, convergence
$p(k \mid y)$	terminal sum-product quantities	posterior uncertainty in segment count	normalization, prior normalizer
$p(t_p = h \mid y, k)$	forward–backward factorization	marginal distribution of ordered boundary p	normalization over h
$p(b_h = 1 \mid y, k)$	sum over ordered boundaries	probability of any changepoint at h	total mass $k-1$
$\widehat{t}_k^{\mathrm{MAP}}$	max-sum recursion and backtracking	joint MAP partition for fixed k	terminal-score equality
$\mathbb{E}[m(\theta_s)^r \mid y]$	sum over segments covering s	model-averaged segment functional	common marginal-likelihood denominator
Predictive score	observation-family posterior predictive	out-of-sample log score under the stated support rule	holdout, family dispatch, coordinates

the factors c_u, h_u; it is not treated as exposure in the observation likelihood.

7.2 Log-space evaluation and failure handling

All segment and partition scores are stored in log space. Sums in Equations (7)–(8) are evaluated by a stable `logsumexp`. Cumulative sufficient statistics should be centered or scaled when appropriate, and conjugate hyperparameters must lie in their proper domains. When a numerical segment calculation fails to converge, violates support, or lacks the stated error control, it returns a structured failure state. The analysis then follows a declared rule: exclude the segment by assigning $-\infty$, terminate the fit, or continue while labeling the posterior as based on an unverified approximation. It must not silently substitute a different observation model.

Tie rules are deterministic and recorded. In max-sum recursion, the predecessor rule identifies which member of a set of equally scoring partitions is returned. In one-to-one boundary matching, the minimum-distance refinement and a fixed tie order identify the matching. These conventions make repeated calculations reproducible; they do not alter the statistical estimand.

7.3 Computational complexity

Table 3 separates asymptotic operation counts from finite-range timing measurements.

Enumeration of all admissible segments is the principal cost of the exact calculation. Pruning, screening, or a restricted changepoint grid can reduce computation, but may also remove partitions unless the restriction is proved exact for the specified model. Such accelerations are not included in the exactness claims of this paper unless their preservation of the partition sum is established separately.

7.4 Reproducibility records

Each returned fit records the method version, data and configuration hashes when available, observation family and hyperparameters, fitted coordinate range, candidate-segment restrictions, priors, random seeds for restart-based calculations, convergence states, approximation diagnostics, and result identifiers. An executed numerical output is not overwritten: a corrected computation receives a new identifier and a link to the earlier result. This distinction is necessary when an output was genuinely computed but cannot answer the scientific question for which it was initially reported.

Table 3. Computational complexity of the implemented calculations. Empirical timings are reported separately in Section 9.

Component	Time	Working memory	Numerical consideration
Conjugate segment precomputation	$\Theta(n^2)$	$\Theta(n^2)$	cumulative sufficient statistics; invalid segments receive $-\infty$
Nonconjugate segment precomputation	$\Theta(n^2 T_{\mathrm{seg}})$	$\Theta(n^2)$ plus method state	convergence and error diagnostics
Sum-product recursion	$\Theta(k_{\max} n^2)$	$\Theta(k_{\max} n)$ for all message layers	log-sum-exp evaluation
Boundary and moment extraction	$\Theta(k_{\max} n^2)$ for streamed fixed-k summaries	output dependent; $\Theta(k_{\max}^2 n)$ for every ordered-boundary table	common marginal-likelihood normalizer
Max-sum recursion and backtracking	$\Theta(k_{\max} n^2)$	$\Theta(k_{\max} n)$ including backpointers	deterministic predecessor rule
Shared-changepoint pooling	$\Theta(Sn^2 + k_{\max} n^2)$ after sequence-specific segment calculations	sequence arrays or streamed log-score sum	add log marginal likelihoods across sequences
Latent-group template update	$\Theta(GSn^2 + Gk_{\max} n^2)$ per iteration	problem dependent	restart ranking must use the final objective value

8 Experimental design and interpretation of results

The archive contains executed synthetic studies and real-data analyses, but it does not fill every validation condition needed for broad comparative claims. Each result is therefore interpreted only for the estimand, observation model, coordinate system, and reference definition used in its calculation. Planned studies are described separately from executed results.

8.1 Questions addressed by the empirical studies

The empirical analysis addresses five questions.

1. **Posterior computation and uncertainty:** do the exact-recursion identities hold, and are boundary-event posterior probabilities calibrated under repeated simulation from the fitted model class?
2. **Generality across observation models:** can the same partition recursions use segment marginal likelihoods from materially different exponential-family and nonconjugate observation models?
3. **Hierarchical structure:** do shared-changepoint and latent-group formulations recover the structures used to generate their synthetic data?
4. **Numerical approximation:** how do segment log marginal-likelihood discrepancies, runtime, selected segment count, and boundary recovery vary among numerical methods?
5. **Real-data behavior:** what posterior segmentations and uncertainty summaries are obtained for the archived well-log, array-CGH, equity-volatility, and methylation datasets?

The executed studies address these questions in the stated designs. They do not establish universal superiority, external domain accuracy, or asymptotic improvements over the quadratic dynamic-programming complexity.

8.2 Synthetic studies

The single-sequence simulations use known piecewise-constant parameters and therefore permit direct boundary and signal-reconstruction metrics. Boundary estimates are matched to generating changepoints by Definition 6.6 at the stated tolerance. The Gaussian calibration study groups posterior boundary-event

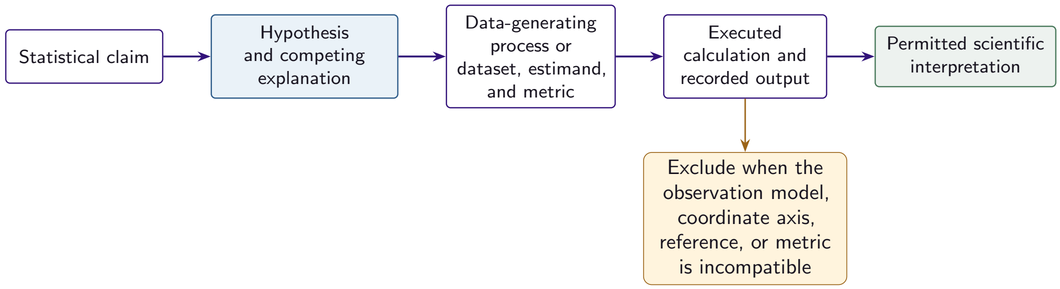


Figure 8. Relation among a statistical claim, the corresponding experiment, the executed output, and the permitted interpretation. Execution of a calculation is distinct from whether that calculation is valid for a particular scientific comparison.

probabilities into bins and reports empirical frequencies, expected calibration error (ECE), and Brier score. The observation-model study uses Gaussian, Poisson, Binomial, and Beta-valued segment models with the same partition recursions; it is a test of the general segment-marginal-likelihood formulation, not a ranking of the four data-generating processes.

The multiple-sequence study with a common changepoint configuration compares pooled and sequence-specific posterior summaries on aligned sequences. The latent-group study uses a finite set of group templates and reports hard allocation accuracy and group-specific boundary marginals in one simulated setting. Because the criterion in Equation (19) is not a normalized finite-mixture likelihood, this study does not test finite-mixture identifiability.

The nonconjugate study compares quadrature with Laplace approximation, the Jaakkola–Jordan bound, Pólya–Gamma mean-field approximation, and expectation propagation. It reports runtime, maximum absolute segment-score discrepancy relative to quadrature, selected segment count, and boundary F_1. These are descriptive comparisons for the archived example. They do not establish a uniform approximation rate for any method.

8.3 Real-data analyses and comparator definitions

The four case studies use the archived preprocessing, priors, segment-count bounds, and observation models. In each plot, red vertical lines denote fitted joint-MAP changepoints unless a separate annotation source is explicitly stated. The current archive does not contain independent changepoint labels for the four applications.

Comparator boundary metrics use the same one-to-one matching rule. When the reference is the BayesBreak joint-MAP boundary set, a comparator score measures agreement with that estimate, not accuracy against independent truth. Tuning and segment-count matching are reported when present. Marginal-likelihood values obtained from different response variables, priors, or factorizations are not interpreted as Bayes factors unless the models define probabilities for the same observed data under a common comparison.

The well-log analysis uses a downsampled sequence from the archived NMR record. The array-CGH analysis uses 43 subjects observed at 2,215 aligned probes. The equity analysis uses the archived 2015–2023 series after stride-four preprocessing. The methylation analysis uses the `methylKit test1.myCpG` chromosome-21 example, not an external methylation-atlas analysis.

8.4 Status of the executed outputs

8.5 Outstanding validation studies

The archive does not yet contain a fairness-controlled external benchmark for every observation model, repeated confidence intervals for all recovery tables, broad misspecification studies, scaling experiments

Table 4. Scientific interpretation of the executed outputs.

Output	Supports	Does not support	Status
Gaussian boundary calibration	calibration in the archived generating design	calibration under arbitrary misspecification	real, admissible
Observation-model study	use of common partition recursions with the stated segment models	a universal performance ranking across families	real, admissible
Latent-group simulation	allocation recovery in one finite-template setting	normalized-mixture identifiability	real, limited to the stated design
Nonconjugate comparison	observed score-error, runtime, and segmentation differences	method-wide approximation rates	real, descriptive
Runtime studies	wall-clock behavior over the measured ranges	asymptotically linear complexity	real, descriptive
Four real-data fits	executed posterior summaries for the archived datasets	external changepoint accuracy or causal event attribution	real, descriptive
Array-CGH flattened comparator	execution of a calculation on incompatible coordinate axes	comparator performance	real, excluded from comparison
Methylation held-out value	execution of the archived fallback calculation	Beta posterior-predictive performance	real, excluded from prediction results

Quantity	Value
Selected k (`k_ml_`)	4
Posterior mean $\mathbb{E}[k]$	4.742
MAP k	4
$\log p(y)$	-18.031

Table 5. Executed posterior summary for Figure 9.

beyond the bundled range, independent real-data changepoint annotations, or a uniform segmentwise error bound for each nonconjugate approximation. These are unresolved empirical or numerical questions. They are stated in Section 10 and are not filled with synthetic placeholder values.

9 Executed results

All numerical values in this section are populated archived outputs. No experiment was rerun for this revision. The figures and tables retain their source sidecars; interpretation follows Section 8.

9.1 Single-sequence posterior summaries

Figure 9 shows a Gaussian sequence with two true changepoints. The boundary-event posterior concentrates near the generating changes while retaining a smaller shoulder of ambiguity before the first jump. The joint-MAP signal reflects this local uncertainty by inserting a short intermediate segment. Table 5 shows that the selected and modal segment count is four while the posterior mean is 4.742, illustrating why an exported partition and posterior uncertainty over k should not be collapsed into one object.

Family	n	$k^\star$	$\hat{k}$	F1@τ	MAE	MSE	$-\log p(y)/n$
Gaussian	120	3	3	0.968	0.00	0.0023	0.191
Poisson	120	3	4	0.777	0.20	0.4501	2.162
Binomial	120	3	3	0.960	0.06	0.0004	2.108
Beta-valued	120	3	3	1.000	0.00	0.0001	2.620

Table 6. Executed synthetic summary across local observation models. The Beta-valued row is quadrature-based rather than closed-form conjugate.

Method	$\max \lvert\Delta \log A^0\rvert$	$q_{95}\lvert\Delta \log A^0\rvert$	median $\lvert\Delta \log A^0\rvert$	time (s)	F1@τ	$\hat{k}$
quadrature	0.000	0.000	0.000	0.093	0.286	6
laplace	1.235	1.188	0.980	0.049	0.286	6
jj	1.265	1.216	1.002	0.071	0.500	3
ep	14.502	14.141	12.680	0.059	0.500	7
pg-vb	1.265	1.216	1.002	0.072	0.500	3

Table 7. Executed local-approximation trade-off. The observed maximum errors are too large to treat this example as verification of a small uniform segmentwise error bound.

9.2 Portability across local observation models

The same global inference code was applied to Gaussian, Poisson, Binomial, and Beta-valued responses (Figure 10). Table 6 reports the archived synthetic metrics. Gaussian, Binomial, and Beta-valued settings selected the generating three segments; Poisson selected four and had the weakest boundary score and largest reconstruction error in this small benchmark. The result supports interface portability and exposes relative difficulty in the bundled settings; it is not a population-level ranking of families.

9.3 Calibration and latent-group segmentation

Repeated Gaussian simulations produced a reliability curve close to the diagonal, with archived ECE approximately 0.010 and Brier score approximately 0.011 (Figure 11a). These values support calibration in the generating design, not robustness to arbitrary misspecification.

The latent-group experiment used 24 sequences of length 80 from two known synthetic groups at noise standard deviation 1.0. It achieved 96% hard assignment accuracy, with one misassignment and visible ambiguity around the class boundary (Figure 11b). Group-specific boundary marginals concentrate near the generating templates. The result is evidence for the finite score-clustering objective in Equation (19); it is not evidence for normalized-mixture identifiability.

9.4 Nonconjugate block approximations

Table 7 compares five local routines in the archived Bernoulli/logistic example, using quadrature as the local reference. Laplace was the fastest listed routine at 0.049 seconds, whereas quadrature required 0.093 seconds. Maximum absolute block-score discrepancies were 1.235 for Laplace, 1.265 for Jaakkola–Jordan and Pólya–Gamma mean field, and 14.502 for expectation propagation. Boundary F_1 did not vary monotonically with block-score discrepancy: quadrature and Laplace selected six segments with $F_1 = 0.286$, whereas three other routines reported $F_1 = 0.500$ under different selected segment counts.

This result demonstrates why segmentation output alone cannot certify a block approximation. The global dynamic program aggregates many local score differences, and a boundary metric can improve despite a

n	$k_{\max}$	mean (s)	std (s)
50	20	0.0455	0.0008
100	20	0.1033	0.0011
200	20	0.2195	0.0011
400	20	0.5106	0.0307
800	20	1.0585	0.0345

Table 8. Archived $k_{\max} = 20$ runtime values from the length sweep.

Table 9. Executed real-data fits. The log evidences belong to different data models and are not compared across rows.

Case	n or shape	$\widehat{k}$	Log evidence
Well-log	507	23	-4989.2835
Array-CGH	43×2215	15	76359.7996
S&P 500	566	29	-1296.6537
Methylation	1904	15	-9518.6675

worse local approximation. Claim-supporting use of Theorem 6.2 therefore requires a uniform error bound over all admissible segments rather than retrospective agreement in the estimated segmentation.

9.5 Finite-range runtime behavior

Figure 12 reports length and segment-budget sweeps. At $k_{\max} = 20$, mean runtime rose from 0.0455 seconds at $n = 50$ to 1.0585 seconds at $n = 800$ (Table 8). Fitted log–log slopes were approximately 1.07 in n for $k_{\max} = 10$, 1.14 for $k_{\max} = 20$, and 0.88 in $k_{\max}$ at fixed $n = 200$. They summarize this implementation over the measured range; they do not replace the $\Theta(k_{\max} n^2)$ worst-case accounting in Proposition 4.4.

9.6 Four real-data fits

Table 9 summarizes the executed case studies. Each fit passed all four archived dynamic-programming checks. Their figures provide fitted segmentation, boundary uncertainty, segment-count posterior information, and in-sample diagnostics. No independent boundary labels were present, so the case studies establish execution and descriptive posterior behavior rather than external accuracy.

Well-log. The archived source contains 4,050 NMR response values and an executed downsampled sequence of $n = 507$. Under the index-uniform prior, the Gaussian fit selected 23 segments with log evidence -4989.283473023698 (Figure 13). A separately executed length-cohesion configuration selected 25 segments with log evidence -4997.147814570523. These are distinct prior models; the values do not establish a universal preference between index and physical-length priors. Archived comparator scores at tolerance three are agreement with the BayesBreak MAP boundary set, not verified geology.

Array-CGH. The shared-boundary model used 43 subjects on 2,215 probes, selected the maximum considered 15 segments, and recorded pooled log evidence 76359.79958695151 (Figure 14). The sum of independently fitted subject evidences was 109617.69544504143, but the two totals arise from different factorizations and are not a direct Bayes factor.

`RES-BB-CMP-002`: **incompatible comparator axis**

> The comparator cache flattened a 43×2215 fitted matrix to 95,245 values while retaining reference boundaries on the 2,215-probe axis. Its zero agreement scores are real executions of an incompatible comparison and are excluded from scientific comparison. A correction requires the raw subject-by-probe matrix, validated probe coordinates, and a new result identifier.

S&P 500 volatility. The source covers January 1, 2015 through December 31, 2023, with approximately 2,263 raw observations and an executed stride-four series of $n = 566$. The Gaussian log-squared-return model selected 29 segments with log evidence -1296.6537252802884 (Figure 15). A secondary Bernoulli threshold-incidence signal selected 50 segments with log evidence -114.41004640965144. It is a different response model, not a Poisson count, and its evidence is not directly comparable with the Gaussian total.

Methylation. The executed data are the `methylKit test1.myCpG` chromosome-21 example with 1,904 CpGs and per-CpG read coverage used as known precision. The Beta-valued quadrature model selected 15 segments and recorded log evidence -9518.667508691196 (Figure 16). This is not the previously described methylation-atlas analysis. The segmentation fit remains admissible descriptive evidence; the archived held-out value is excluded from predictive conclusions under Definition 6.8.

9.7 What the executed record establishes

The archive demonstrates that common partition recursions can use closed-form conjugate segment marginal likelihoods, one-dimensional quadrature, other numerical segment calculations, shared-changepoint pooling, and finite latent-group templates. It provides calibrated boundary probabilities in one Gaussian simulation design, finite-range runtime measurements, and four traceable real-data fits. It does not yet establish external boundary accuracy, dominance over tuned comparators, robustness across broad misspecification regimes, routine-wide approximation guarantees, or asymptotic speed better than the quadratic dynamic-programming bound.

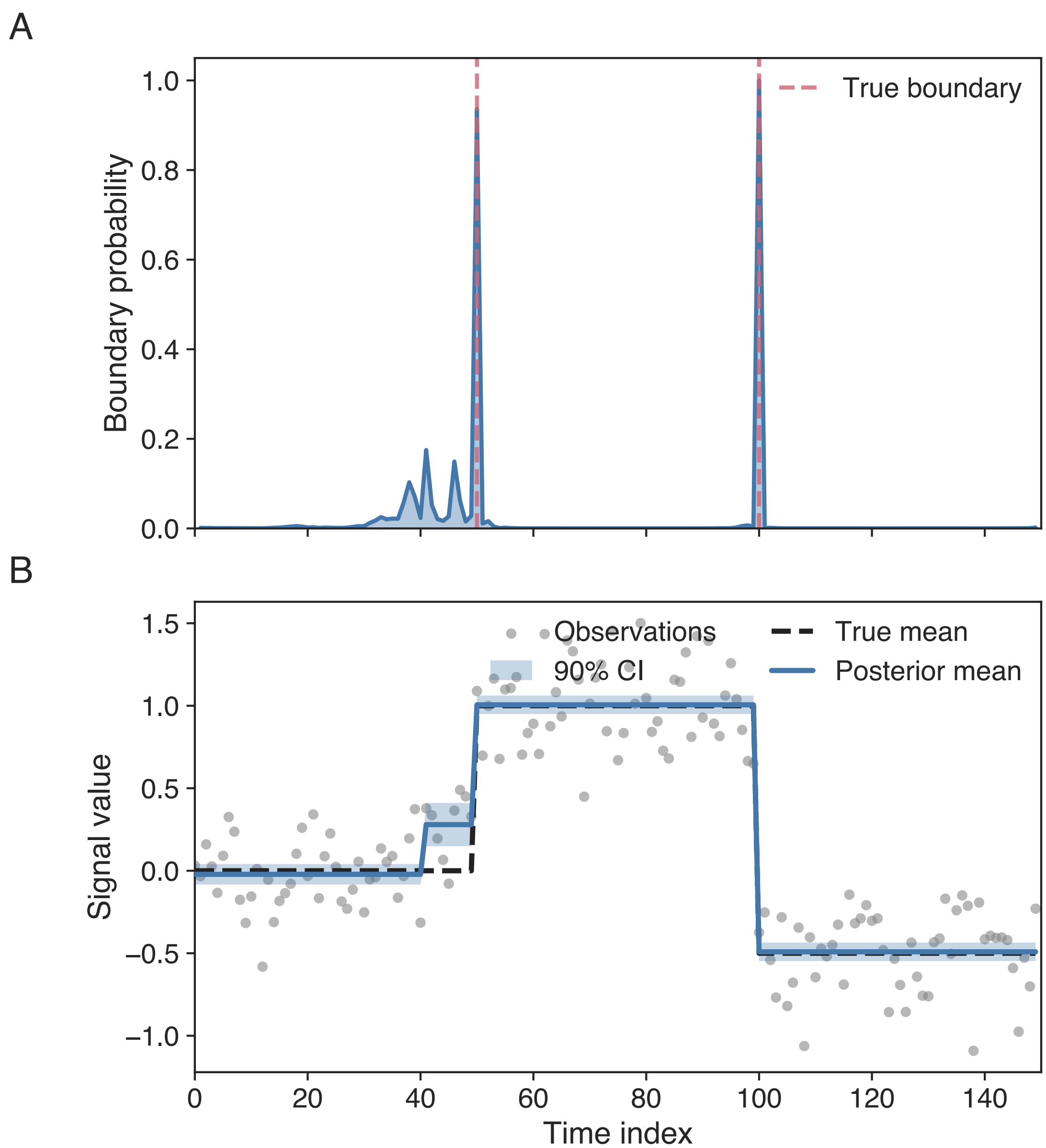


Figure 9. Executed Gaussian illustration. Top: marginal boundary-event posterior with generating changepoints shown by dashed lines. Bottom: observations, latent signal, joint-MAP segmentation, and 90% segment-level posterior intervals.

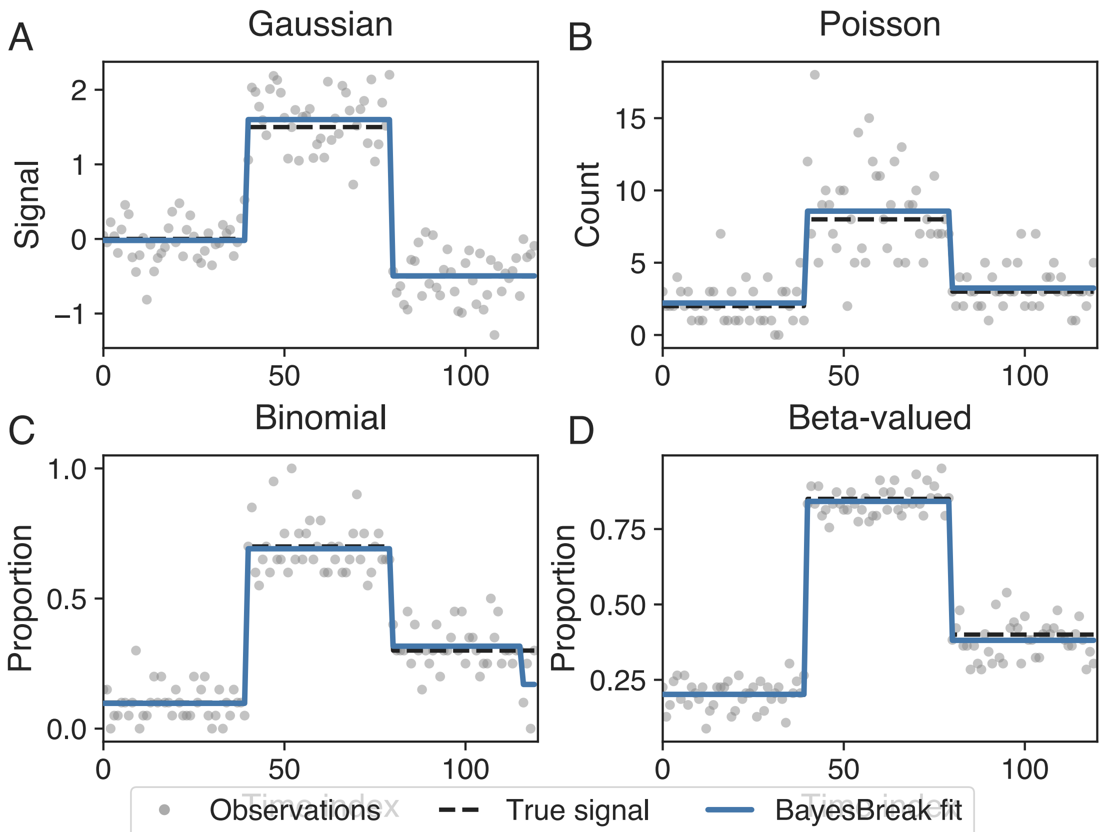
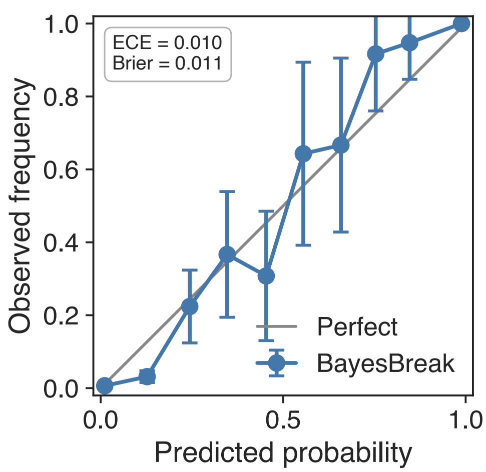
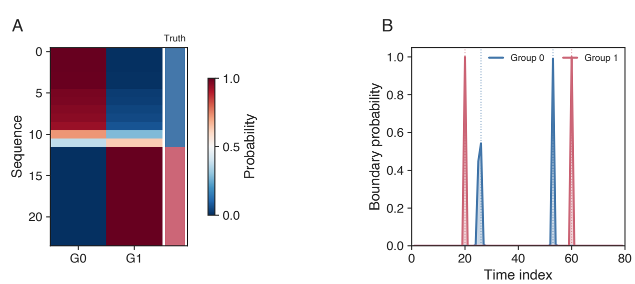


Figure 10. Executed family showcase. Gaussian, Poisson, and Binomial use closed-form local conjugate integration; the Beta-valued response uses deterministic one-dimensional quadrature.



(a) Boundary-event reliability.



(b) Latent-group assignments and group-specific boundary marginals.

Figure 11. Executed uncertainty and template diagnostics. Panel (a) reports ECE ≈ 0.010 and Brier ≈ 0.011 in the archived Gaussian calibration experiment. Panel (b) reports 96% hard recovery for the declared two-template synthetic setting.

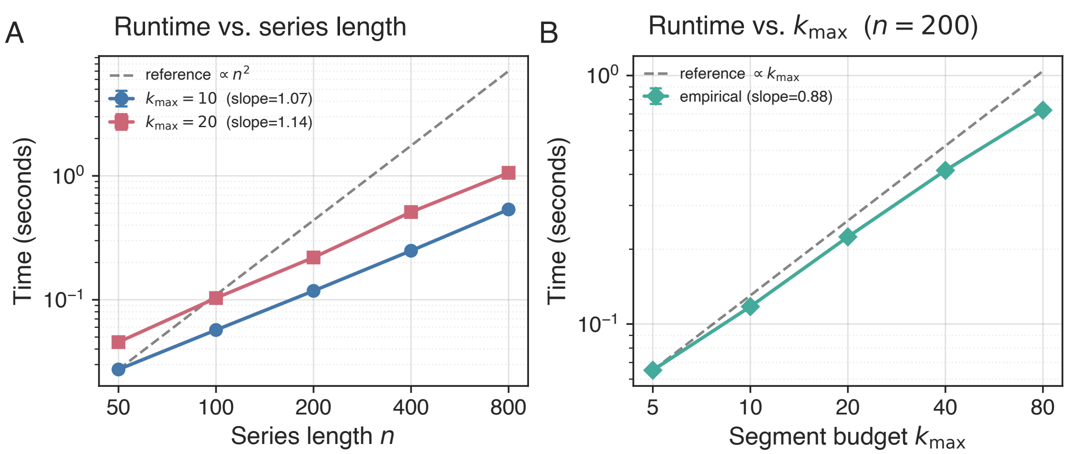


Figure 12. Executed Gaussian runtime study. Points are means over five repetitions and error bars are one standard deviation. Dashed references show quadratic-in-n and linear-in-$k_{\max}$ guides, not fitted complexity claims.

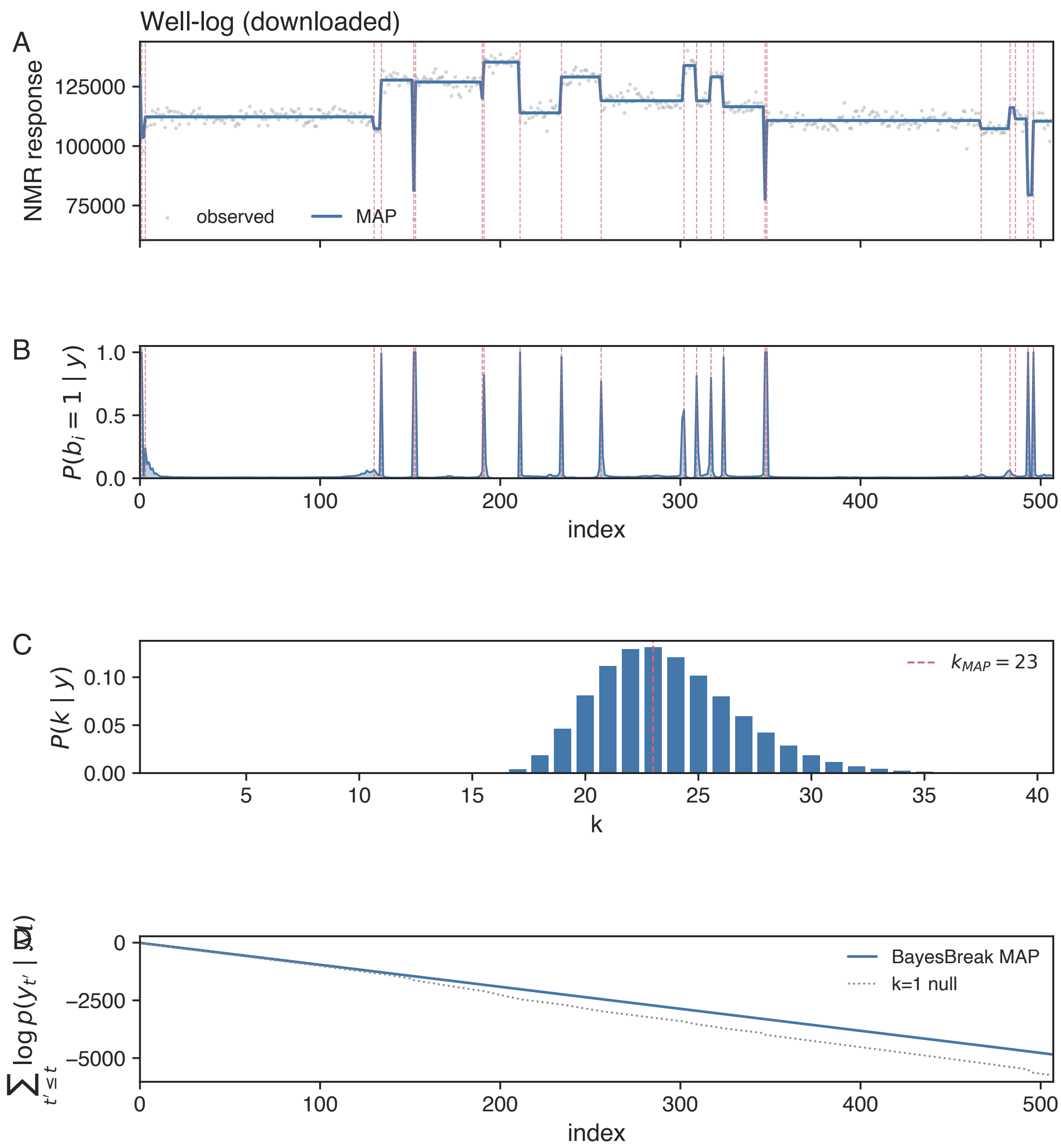


Figure 13. Executed well-log fit. Solid segmentation and red markers are fitted summaries; boundary probabilities and $p(k \mid y)$ are posterior quantities; the cumulative panel is in-sample.

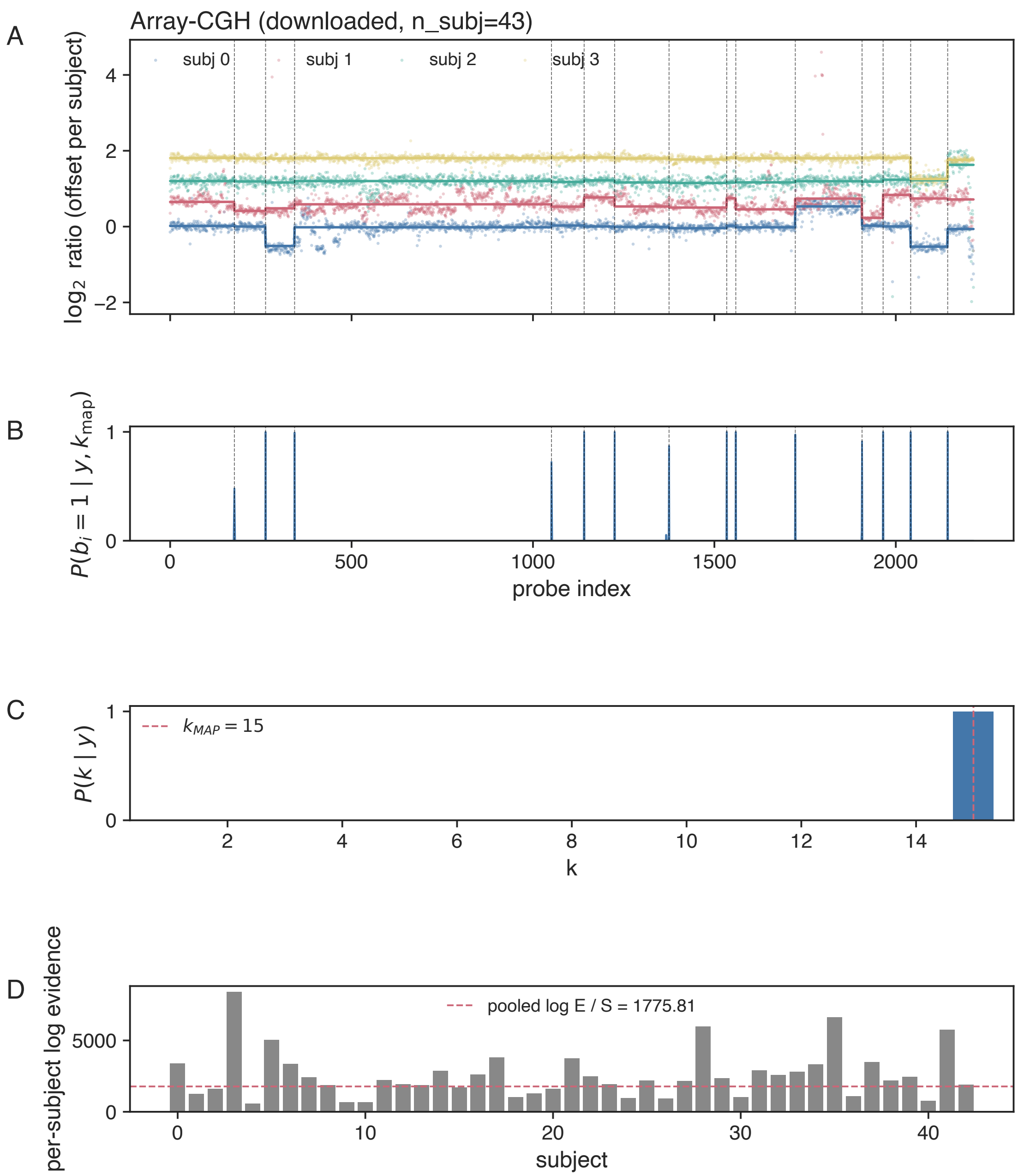


Figure 14. Executed shared-boundary array-CGH fit. The common red markers are fitted joint-MAP boundaries, not external copy-number annotations.

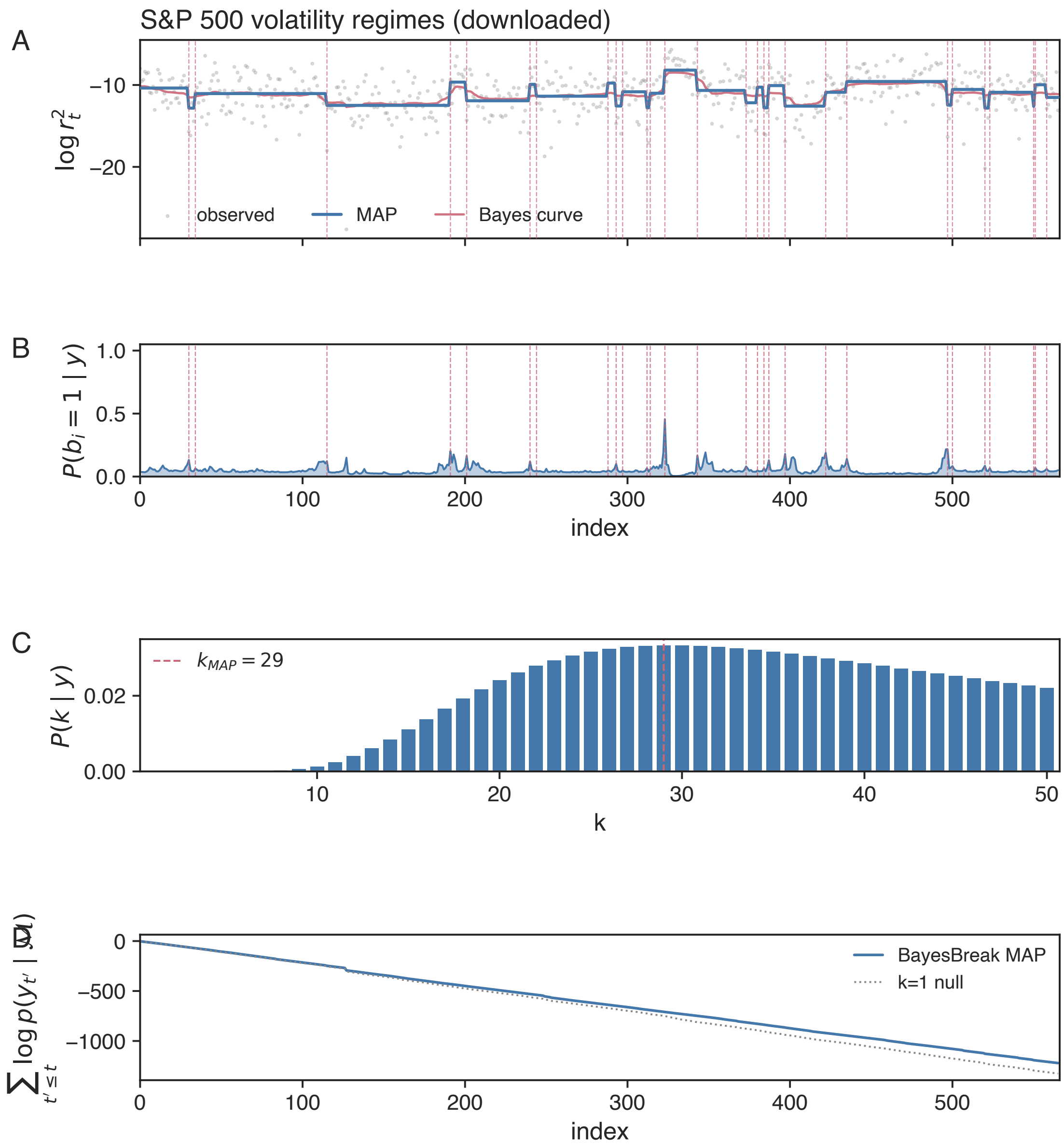


Figure 15. Executed S&P 500 volatility fit after archived stride-four preprocessing. Red markers are fitted boundaries, not independently annotated macroeconomic events.

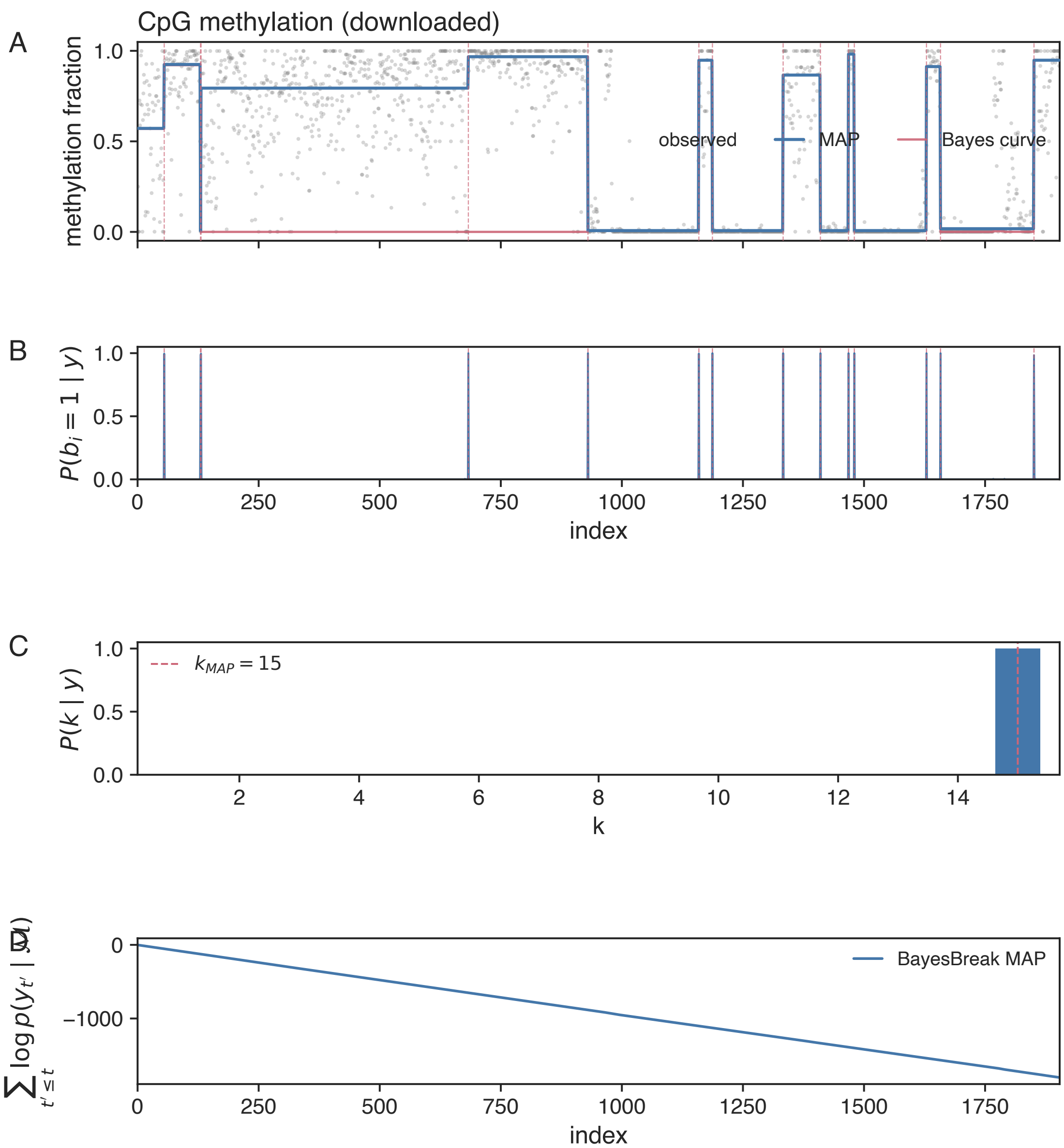


Figure 16. Executed Beta-valued fit to the methylKit chromosome-21 example. Red markers are fitted MAP boundaries; no independent atlas transition annotation was archived.

10 Limitations and scope

The partition recursions are exact for the stated offline, contiguous, locally factorized segmentation model when each segment marginal likelihood is exact. The following conditions delimit the method and the empirical conclusions.

Computational regime. The exact dynamic program has worst-case cost $\Theta(k_{\max} n^2)$ and is not an online changepoint detector. It is practical for moderate sequence lengths and segment-count bounds. Larger problems require a restricted candidate set, a pruning rule with its own correctness result, or an explicitly approximate calculation. The finite-range timing slopes in Section 9 do not alter the worst-case bound.

Candidate changepoints and dependence across segments. A changepoint absent from the candidate set cannot be recovered. Nonlocal interactions among segments, unrestricted dependence among segment labels, and priors that cannot be represented by local transition factors fall outside the dynamic program derived here. Some forms of dependence can be incorporated by enlarging the state space, but that produces a different recursion and requires a separate analysis.

Observation-model misspecification. Posterior conclusions are conditional on the chosen segment model, descriptor semantics, hyperparameters, and within-segment independence assumptions. Heavy tails under a Gaussian model, zero inflation under a Poisson model, overdispersion under a Binomial model, and unmodeled serial dependence can distort segment marginal likelihoods and hence the posterior over partitions. The method does not by itself identify the scientifically appropriate observation family.

Prior sensitivity. Posterior distributions for segment count and changepoint locations can be sensitive to $p(k)$, $c_u(i, j)$, and $h_u(j)$, especially with irregularly spaced observations. Sensitivity to segment cohesion and boundary hazard should be reported. Marginal-likelihood values from analyses with different priors describe different models and should not be interpreted as a single Bayes-factor comparison without specifying the compared probability models.

Nonconjugate approximation. Theorem 6.2 is conditional on Assumption 6.1. The effect of a fixed segmentwise log-score error increases with the number of segments. A numerical method that does not establish the uniform error bound yields an exact dynamic-programming calculation for its approximate score matrix, not an error-controlled approximation to the intended posterior. In the archived study, expectation propagation had maximum absolute segment-score discrepancy about 14.5, which is not a small-error setting.

Latent-group non-uniqueness. The finite latent-group criterion is invariant to permutation of group labels and may admit duplicate or collapsed templates. Multiple restarts do not guarantee a global maximum. The archived implementation can return a stale objective value along one convergence path; that defect must be corrected before objective values are used to rank restarts. The present analysis does not imply identifiability of a normalized finite-mixture model.

Shared-changepoint assumptions. Exact pooling assumes aligned candidate coordinates and conditional independence of sequences given a common partition. Dependence among sequences or genuinely sequence-specific changepoints can make a common partition inappropriate. The finite-candidate consistency result additionally requires a positive average expected log-score gap; information from only one sequence is not sufficient as the number of sequences grows.

Posterior prediction. The observation family and coordinate support are part of the predictive model. A fallback to another family or silent assignment of out-of-range coordinates can produce a finite but invalid predictive score, as shown by the excluded methylation computation. Prediction outside the fitted coordinate range requires an explicit extrapolation model.

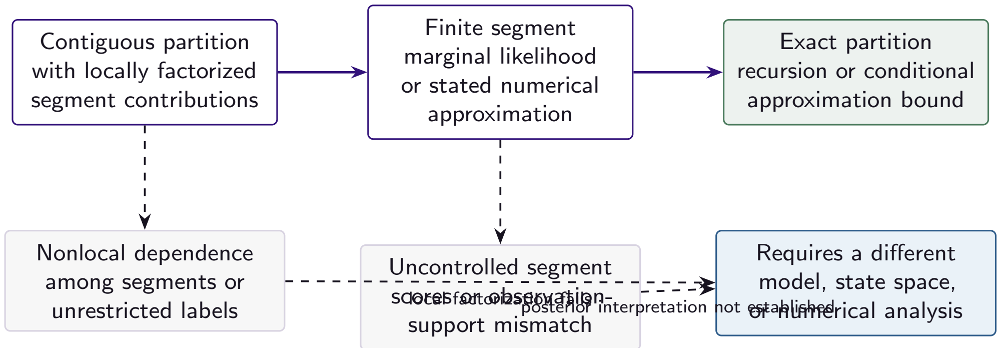


Figure 17. Scope of the exact partition calculation. The recursion cannot compensate for a missing candidate changepoint, nonlocal segment dependence, a misspecified observation model, uncontrolled numerical segment scores, an undefined extrapolation rule, or comparator outputs on incompatible coordinate axes.

Evaluation of changepoints. Boundary F_1 depends on the tolerance and matching rule. Agreement with the BayesBreak joint-MAP set is not accuracy against independent truth. Matched location error is undefined when no pair is matched. Red lines in the case-study plots are fitted changepoints rather than external annotations. The archived analyses therefore do not support causal attribution to geological, genomic, or financial events.

Incomplete empirical coverage. The current archive lacks broad repeated uncertainty intervals, a fully fairness-controlled comparator study, comprehensive misspecification grids, larger scaling studies, and method-specific error bounds for the nonconjugate approximations. These omissions limit comparative, robustness, and approximation claims even though the populated numerical values are real executed results.

Implementation verification. The historical Phase 1 bounded verification collected 179 tests: 157 passed, five were skipped because of optional dependencies, and 17 nonconjugate or grouped-model tests did not finish. An independent Phase 6 rerun collected the same 179 tests: 173 passed, five were skipped, one EP logistic-normal test exceeded a 20-second per-test cap, and none failed. The remaining timeout is neither a pass nor a failure. No package code or research experiment was changed or rerun for this verification.

BayesBreak is therefore a generalized hierarchical Bayesian method for offline segmentation of ordered data under locally factorized segment models. It is not a universal changepoint procedure. Online detection, changes in slope, reversible-jump posterior simulation, and segmentation of nonordered graphs require different models or additional theory (Adams & MacKay, 2007; Ruggieri & Lawrence, 2014; Green, 1995; Fearnhead, 2006).

11 Conclusion

We have developed a generalized hierarchical Bayesian method for multiple-changepoint segmentation with irregular designs, multiple aligned sequences, and known or latent group structure. For regular exponential-family segment models with proper conjugate priors, sufficient statistics give exact segment marginal likelihoods and posterior moments. These quantities enter common dynamic-programming recursions for the posterior distribution over segment counts and changepoint locations. A separate max-sum recursion with backtracking returns the joint MAP partition, so marginal posterior summaries and joint optimization are not conflated.

The same partition formulation accommodates design-dependent segment cohesion and boundary hazards, exact pooling of conditionally independent sequences with common changepoints, and a finite latent-group criterion optimized by alternating minorization and maximization. Nonconjugate segment models can be incorporated through numerical marginal-likelihood calculations; the resulting posterior approximation is related to the intended posterior only when a segmentwise error bound such as Assumption 6.1 is established. Posterior prediction is likewise defined by the fitted observation family and its coordinate support.

The archived studies show that the implementation applies the same partition recursions to several observation models, produces calibrated Gaussian boundary probabilities in one repeated-simulation design, achieves 96% hard allocation accuracy in one finite latent-group simulation, and fits four heterogeneous real datasets. The measured runtime increases from 0.0455 to 1.0585 seconds over the archived $k_{\max} = 20$ length study. These results are real but limited to their stated designs. The archive does not provide independent changepoint annotations for the applications, method-wide nonconjugate error rates, or a complete fairness-controlled benchmark; two executed calculations are excluded because their coordinate or predictive definitions do not match the reported scientific quantity.

The contribution is therefore the generalized statistical formulation and its hierarchical extensions: exponential-family segment integration, exact posterior computation over ordered partitions, irregular-design priors, multiple-sequence pooling, and grouped or latent-group segmentation within one Bayesian method. The claims are grounded in the probability model, the dynamic-programming identities, the stated optimization objective, and the executed experiments rather than in new terminology for established statistical operations.

Acknowledgements and declarations

The archival materials identify Omid Shams Solari of sAIm Labs as the author and corresponding contact. Funding, competing-interest, ethics, and contributor-role declarations were not present in the supplied source package and must be completed by the author before journal submission.

The source archive contains the implementation, tests, experiment scripts, result sidecars, figures, tables, and provenance records used in this manuscript. A public repository or permanent data-release URL was not supplied; availability language should therefore be updated only after the author selects a release location. No package code was changed and no research experiment was rerun during this manuscript revision.

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Appendix

A Local block formula reference

This appendix records the principal family instantiations used by the executed archive. They are direct substitutions into Theorem 3.2; family descriptors remain distinct from optional likelihood powers.

A.1 Gaussian mean with known observation variance

For $y_r \mid \mu \sim \mathcal{N}(\mu, \sigma^2/w_r)$ and $\mu \sim \mathcal{N}(\nu, \rho^2)$, let

$$W_{ij} = \sum_{r=i+1}^{j} w_r, \qquad S_{ij} = \sum_{r=i+1}^{j} w_r y_r,$$
$$Q_{ij} = \sum_{r=i+1}^{j} w_r (y_r - \nu)^2, \qquad \kappa_{ij} = \rho^2 W_{ij} + \sigma^2.$$

The posterior mean and variance are

$$\nu_{ij} = \frac{\rho^2 S_{ij} + \sigma^2 \nu}{\kappa_{ij}}, \qquad \rho_{ij}^2 = \frac{\sigma^2 \rho^2}{\kappa_{ij}},$$

and the integrated block log evidence is

$$\begin{aligned}\log A_{ij}^{(0)} = &-\frac{j-i}{2}\log(2\pi\sigma^2) + \frac{1}{2}\sum_{r=i+1}^{j}\log w_r \\ &- \frac{1}{2}\log\left(1 + \frac{\rho^2 W_{ij}}{\sigma^2}\right) - \frac{Q_{ij}}{2\sigma^2} \\ &+ \frac{(S_{ij} - \nu W_{ij})^2}{2(\sigma^2 W_{ij} + \sigma^4/\rho^2)}. \end{aligned} \tag{28}$$

The first two moment numerators multiply this evidence by ν_{ij} and $\rho_{ij}^2 + \nu_{ij}^2$.

A.2 Poisson counts with exposure

For $y_r \mid \lambda \sim \mathrm{Pois}(\lambda e_r)$ and $\lambda \sim \mathrm{Gamma}(a_0, b_0)$ in shape–rate form, define $C_{ij} = \sum y_r$, $E_{ij} = \sum e_r$, and $H_{ij} = \sum [y_r \log e_r - \log(y_r!)]$ over the block. Then

$$A_{ij}^{(0)} = e^{H_{ij}} \frac{b_0^{a_0} \Gamma(a_0 + C_{ij})}{\Gamma(a_0)(b_0 + E_{ij})^{a_0 + C_{ij}}}, \tag{29}$$

and the posterior is $\mathrm{Gamma}(a_0 + C_{ij}, b_0 + E_{ij})$.

A.3 Binomial observations with known trials

For $y_r \mid p \sim \mathrm{Binom}(m_r, p)$ and $p \sim \mathrm{Beta}(a_0, b_0)$, define $C_{ij} = \sum y_r$, $M_{ij} = \sum m_r$, and $H_{ij} = \sum \log \binom{m_r}{y_r}$. Then

$$A_{ij}^{(0)} = e^{H_{ij}} \frac{B(a_0 + C_{ij}, b_0 + M_{ij} - C_{ij})}{B(a_0, b_0)}, \tag{30}$$

and the posterior is $\mathrm{Beta}(a_0 + C_{ij}, b_0 + M_{ij} - C_{ij})$.

A.4 Beta-valued observations

For $y_r \in (0, 1)$ with known precision ϕ_r and block mean $\mu \in (0, 1)$,

$$y_r \mid \mu, \phi_r \sim \mathrm{Beta}(\phi_r \mu, \phi_r (1 - \mu)),$$
$$\mu \sim \mathrm{Beta}(a_0, b_0).$$

This pair is not conjugate in μ. Write $g_s(\mu) = \mathrm{BetaPDF}(y_s \mid \phi_s \mu, \phi_s(1 - \mu))$ and $\pi_0(\mu) = \mathrm{BetaPDF}(\mu \mid a_0, b_0)$. The local interface uses deterministic one-dimensional quadrature,

$$A_{ij}^{(r)} = \int_0^1 \mu^r \exp\left\{ \sum_{s=i+1}^{j} \log g_s(\mu) + \log \pi_0(\mu) \right\} \mathrm{d}\mu. \tag{31}$$

Node-count sensitivity and endpoint behavior are numerical diagnostics; this integral is not represented as a closed-form conjugate result.

B Log-space dynamic programs

Algorithm 2: Log-space sum-product messages

Input: Prior-adjusted log scores $\widetilde{\ell}_{ij}$; $k_{\max}$.

- Initialize $\Lambda_{0,0} = 0$, $\Lambda_{0,j>0} = -\infty$, $\mathcal{R}_{0,n} = 0$, and $\mathcal{R}_{0,i<n} = -\infty$;
- **for** $k = 0$ **to** $k_{\max} - 1$ **do**
  - **for** $j = k + 1$ **to** n **do**
    - $\Lambda_{k+1,j} \leftarrow \mathrm{LSE}_{h=k}^{j-1}\{\Lambda_{k,h} + \widetilde{\ell}_{h,j}\}$;
  - **for** $i \leftarrow n - k - 1, n - k - 2, \ldots, 0$ **do**
    - $\mathcal{R}_{k+1,i} \leftarrow \mathrm{LSE}_{h=i+1}^{n-k}\{\widetilde{\ell}_{i,h} + \mathcal{R}_{k,h}\}$;
- Normalize $\log p(k) + \Lambda_{k,n} - \log C_k(u)$ over k;

Output: Λ, $\mathcal{R}$, and $p(k \mid y)$.

Algorithm 3: Joint-MAP max-sum and backtracking

Input: Prior-adjusted log scores $\widetilde{\ell}_{ij}$; fixed k.

- Set $M_{0,0} = 0$ and $M_{0,j>0} = -\infty$;
- **for** $q = 1$ **to** k **do**
  - **for** $j = q$ **to** n **do**
    - $h^\star(q, j) \leftarrow \arg\max_{h=q-1}^{j-1}\{M_{q-1,h} + \widetilde{\ell}_{h,j}\}$ under the declared tie rule;
    - $M_{q,j} \leftarrow M_{q-1,h^\star(q,j)} + \widetilde{\ell}_{h^\star(q,j),j}$;
- Set $\widehat{t}_k = n$ and recursively $\widehat{t}_{q-1} = h^\star(q, \widehat{t}_q)$;

Output: $\widehat{t}$ and terminal score $M_{k,n}$.

The recommended numerical checks are

$$\sum_{k=1}^{k_{\max}} p(k \mid y) = 1,$$
$$|\Lambda_{k,n} - \mathcal{R}_{k,0}| \leq \delta,$$
$$\sum_{h=1}^{n-1} p(b_h = 1 \mid y, k) = k - 1,$$
$$\left| M_{k,n} - \sum_{q=1}^{k} \widetilde{\ell}_{\widehat{t}_{q-1}, \widehat{t}_q} \right| \leq \delta.$$

with a documented floating-point tolerance δ.

C Supplementary agreement diagnostics

The tables below preserve executed comparator values while making the reference semantics explicit. They are not external-accuracy tables.

Table 10. Well-log boundary agreement with the BayesBreak MAP boundary set at tolerance three. No independent geology annotation was archived.

Method	Agreement F_1@3	Interpretation
Fearnhead-configured reference DP	0.93	agreement with BB MAP
Optimal partitioning	0.91	agreement with BB MAP
Binary segmentation	0.86	agreement with BB MAP
PELT	0.18	agreement with BB MAP
Wild binary segmentation	0.05	agreement with BB MAP

Table 11. S&P 500 and methylation comparator agreement. Segment counts were matched where stated by the archived protocol.

Dataset	Method	$\widehat{k}$	Agreement $F_1@3$	Scope
S&P 500	optimal partitioning, matched k	29	0.82	vs BB Gaussian MAP
S&P 500	PELT	79	0.53	vs BB Gaussian MAP
S&P 500	binary segmentation, matched k	29	0.43	vs BB Gaussian MAP
S&P 500	wild binary segmentation, matched k	29	0.07	vs BB Gaussian MAP
S&P 500	Fearnhead-configured reference DP	13	0.55	vs BB Gaussian MAP
Methylation	optimal partitioning, matched k	15	0.79	vs BB Beta-valued MAP
Methylation	binary segmentation, matched k	15	0.57	vs BB Beta-valued MAP
Methylation	PELT	5	0.44	vs BB Beta-valued MAP
Methylation	wild binary segmentation, matched k	15	0.14	vs BB Beta-valued MAP

The incompatible array-CGH comparator output is intentionally absent from these tables. Its flattened 95,245-value axis and the 2,215-probe boundary axis do not define a common matching problem.

D Result provenance and interpretation

Table 12. Principal executed-result ledger used by the paper. "Real" records execution provenance; the final column states whether and how the output may support the stated scientific target.

Record	Computation	Execution status	Scientific use	Interpretation status
Synthetic Gaussian	posterior and MAP summaries	real	estimator separation, recovery illustration	valid for the stated target
Family showcase	four local block models	real	interface portability	valid for the stated target, scoped
Boundary calibration	repeated Gaussian simulation	real	ECE/Brier in generating design	valid for the stated target, scoped
Latent-group templates	24 labeled sequences	real	finite latent-group allocation behavior	partially scoped
Nonconjugate table	five block routines	real	descriptive trade-off	uncertified globally
Runtime sweep	n and $k_{\max}$ sweeps	real	finite-range performance	descriptive
Well-log	Gaussian case-study fit	real	descriptive segmentation	descriptive only; no reference truth
Array-CGH	shared-boundary fit	real	descriptive pooled fit	descriptive only; no reference truth
`RES-BB-CMP-002`	flattened CGH comparator	real	negative control for schema checks	excluded from the stated analysis
S&P 500	Gaussian and Bernoulli fits	real	descriptive regime summaries	descriptive only; no reference truth
Methylation segmentation	Beta-valued fit	real	descriptive CpG segmentation	descriptive only; no reference truth
`RES-BB-RD-007Q`	held-out methylation value	real	negative control for family/support checks	excluded from the stated analysis

A corrected computation does not replace an excluded archived row. It receives a new identifier, records its parent, and states the corrected family, axis, support, configuration, and output hashes.

E Reproducibility checklist

A claim-supporting run should archive the following fields:

1. input data identity, preprocessing, coordinate system, split membership, and hash;
2. observation family, descriptors, likelihood powers, hyperparameters, and support checks;
3. segment-count prior, cohesion, hazard, candidate-block constraints, and $k_{\max}$;
4. block routine, software version, convergence state, approximation reference, and error certificate when applicable;
5. forward/backward and max-sum invariant checks with numerical tolerances;
6. random seeds and complete objective traces for restart-based extensions;
7. prediction family dispatch, fitted coordinate support, and extrapolation rule;
8. metric tolerance, one-to-one matching implementation, and undefined-value policy;
9. result identifier, lineage, configuration hash, artifact paths, and figure/table generator;
10. package test summary with passes, skips, unresolved cases, and failures reported separately.

The historical verification record `RES-BB-QA-002` contains 179 collected tests, 157 passes, five dependency-based skips, and 17 unresolved cases. The independent Phase 6 record `RES-BB-QA-003` contains 179 collected tests, 173 passes, five skips, one timeout under a 20-second per-test cap, and zero failures. Document compilation is not counted as package-level scientific validation.